

DisMUSE: An Efficient Distributed Singular-Spectrum-Based Optimization Approach with Theoretical Convergence Guarantee

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Abstract

In recent years, singular-spectrum-based optimizers (e.g., Muon, SOAP, and Shampoo) have attracted increasing attention due to their ability to preserve the matrix structure of neural network gradients, thereby enabling geometry-aware optimization. So far, however, most singular-spectrum-based (SSB) approaches have remained limited to the centralized setting, which is ill-suited to modern distributed machine learning infrastructures that exploit massive parallelism to accelerate training. Distributed singular-spectrum-based optimizers are still in their infancy, and many open problems remain (e.g., theoretical convergence guarantees and communication efficiency). This gap motivates us to investigate efficient distributed SSB optimizers with theoretical finite-time convergence rate guarantees. Toward this end, we propose the Distributed Momentum-based Spectral Estimator (DisMUSE) method, a principled singular-spectral optimization framework for distributed learning, which covers both data-center and federated settings. Using Muon-style exact orthogonalized momentum as the spectral update, we integrate a recursive variance-reduced momentum estimator and establish an $O((KT)^{-1/3})$ convergence rate, where K denotes the number of nodes and T represents the number of communication rounds. This result matches the best-known rates for variance-reduced nonconvex distributed optimization. Building on this variance-reduced distributed SSBO backbone, we further study two practical extensions. First, we show that replacing exact orthogonalization with finite-step Newton–Schulz (NS) iterations preserves the same convergence order up to a multiplicative approximation factor that rapidly approaches 1 as the number of iterations increases. Second, we develop a communication-efficient variant based on low-rank momentum transmission and characterize the resulting accuracy–communication trade-off through an explicit additive bias term.

CCS Concepts

• Theory of computation → Distributed algorithms.

Keywords

Distributed, Nonconvex, Federated Optimization

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1 Introduction

Distributed learning has become a core paradigm for modern machine learning systems, spanning data-parallel training, federated learning, and edge learning scenarios where computation and data are spread across multiple nodes. In such settings, optimization algorithms must handle stochastic gradient noise, statistical heterogeneity, and communication constraints. While a large body of prior work studies these issues for conventional vector-based methods such as SGD [9] and Adam [20], these methods treat parameters as flat vectors. Much less is understood when parameters and gradients are naturally matrix-valued, as in Transformers and convolutional networks.

Recently, Singular-Spectrum-Based Optimization (SSBO) methods, such as Muon [16], Shampoo [13], and SOAP [47], have attracted growing attention for their ability to preserve the matrix structure of neural network updates. Among them, Muon has garnered the most interest due to its exceptional efficiency in large language model (LLM) training. Muon also has a relatively simple implementation: it orthogonalizes a matrix-valued momentum before applying the update, yielding geometry-aware descent directions [15, 27]. However, despite the promise of these SSBO methods, both their theory and implementation remain largely confined to centralized training, leaving open the question of how they should be designed and analyzed in distributed settings.

However, extending SSBO-based methods to distributed learning is highly non-trivial due to its several technical challenges.

- **Nonlinearity:** Unlike standard vector-based methods, SSBO-based methods apply a *nonlinear* spectral mapping (e.g., orthogonalization) to a matrix-valued gradient or momentum estimator. This nonlinear mapping breaks *commutativity*, a highly useful structural property that is crucial in distributed optimization algorithm design. Specifically, in distributed SSBO-based algorithm design, averaging locally transformed spectral directions from multiple nodes is generally *not* equivalent to applying the spectral map after aggregating the underlying momentum estimates from all data samples as in their centralized counterpart. As a result, directly using local spectral updates in a distributed aggregation typically introduces a systematic bias.
- **Data Heterogeneity:** The nonlinearity challenge is further exacerbated under data heterogeneity scenarios, which often arise

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in federated-type settings. This fundamental nonlinearity challenge suggests that, in SSBO-based optimization algorithms for distributed learning, careful design is required to synchronize and control the matrix-valued estimator prior to applying the nonlinear spectral map, rather than naively averaging locally orthogonalized directions.

- **Noisy Gradient Estimates Coupled with Nonlinearity:** In distributed stochastic training, each node only observes noisy gradient estimates. Unfortunately, unlike standard stochastic gradient-descent-type methods, where such noise can be averaged out through linear aggregation, the stochastic deviations in the momentum estimators are coupled with the nonlinear spectral mapping in SSBO-based methods, and these deviations can be transformed and potentially amplified in a complex way into directional errors by the nonlinear spectral transformation.
- **Matrix-Valued Momentum:** In SSBO-based methods, exact orthogonalization is computationally expensive and is typically replaced in practice by finite-step approximations such as Newton-Schulz (NS) iterations, whose approximation error must be explicitly accounted for in the analysis. However, theoretical understanding of distributed SSBO-based methods remains very limited in the literature. Moreover, synchronizing full matrix-valued momentum variables incurs substantial communication cost, which typically necessitates low-rank approximation approaches for distributed SSBO-based methods. However, naively transmitting low-rank spectral factors can introduce additional distortion if aggregation is not treated judiciously in distributed SSBO-based algorithmic designs.

The main contribution of this work is that we propose a series of new SSBO-based algorithmic techniques to overcome the aforementioned challenges and achieve low sample and communication complexities for distributed SSBO-based learning. In this work, we focus on Muon-type SSBO algorithmic framework due to its recent success in LLM training. Toward this end, we propose the *Distributed Momentum-based Spectral Estimator* (DisMUSE) method, a unified framework for distributed SSBO. The key idea of DisMUSE is to build a variance-reduced distributed SSBO backbone, which is used as the foundation for practical Newton-Schulz orthogonalization and communication-efficient low-rank synchronization. Our contributions are summarized as follows:

- **A variance-reduced Distributed SSBO Backbone:** We first develop a basic version of DisMUSE with exact orthogonalization and full communication of matrix-valued momentum. Here, a STORM-style recursive estimator controls the stochastic deviation of the matrix-valued momentum before the nonlinear spectral transformation. Under standard smoothness and heterogeneity assumptions, we prove that this basic version of DisMUSE achieves an $O((KT)^{-1/3})$ convergence rate for averaged stationarity, matching the best-known rates for variance-reduced nonconvex distributed optimization.
- **An Efficient Implementation of DisMUSE with Finite-step Newton-Schulz Orthogonalization:** Building upon the above basic version of DisMUSE with variance-reduced distributed SSBO, we replace the exact orthogonalization step with a finite number of Newton-Schulz (NS) iterations and show that the

same $O((KT)^{-1/3})$ convergence order is preserved up to a multiplicative approximation factor, which rapidly approaches 1 as the number of NS steps increases.

- **A Communication-efficient Low-rank Implementation:** We further extend the basic version of DisMUSE with full matrix-valued momentum communication to a low-rank communication-efficient implementation, where nodes can transmit low-rank momentum factors instead of full matrices. We explicitly characterize the resulting communication-accuracy trade-off through an additive bias term, which is shown to be determined by the spectral tail of the update matrices.

2 Related Work

In this section, we provide an overview on three closely related research areas to put our work into comparative perspectives.

1) Singular-Spectrum-Based Optimization and Muon: Recent progress in large-scale neural network training has led to a class of optimization methods that explicitly exploit the matrix structure of model updates by manipulating their singular spectra, rather than applying element-wise preconditioning, such as Shampoo [13], SOAP [47], and Muon [16]. These methods have demonstrated strong empirical performance on a variety of deep learning tasks [2, 5, 16, 27, 34, 39, 49, 50]. In the literature, these methods are sometimes loosely referred to as “matrix-aware optimization” or “geometric optimization.” We note, however, that there also exist “matrix-aware methods” that respect tensor structure but do not necessarily operate on spectral properties (e.g., Adafactor [40]) or “geometric optimization” that broadly covers manifold-based updates but without isolating spectrum-level control (ranging from stochastic gradient descent to natural gradient methods [1]). To avoid confusion, in this paper, we use the “Singular-Spectrum-Based Optimization” (SSBO) terminology for those methods that directly manipulate the singular spectrum of matrix updates.

As mentioned earlier, among SSBO-based approaches, Muon has garnered increasing attention in recent years. Muon orthogonalizes a momentum matrix before applying the update, thereby aligning the descent direction with the dominant spectral geometry of the matrix-valued gradient [7]. In practice, Muon typically avoids exact singular value decomposition (SVD) by using iterative approximations, such as NS orthogonalization [16]. A growing body of recent work has studied the theoretical behavior of Muon and related spectral optimizers, including their convergence properties, geometric interpretations, and optimization benefits [2, 10, 10, 15, 19, 22–24, 28, 29, 33, 38, 41, 42]. However, most existing works remain limited to centralized training and are based on exact orthogonalization, leaving a substantial gap between theoretical guarantees and practical implementations in distributed environments.

2) Communication-Efficient Distributed Learning: In data center settings, distributed learning accelerates model training by leveraging the massive parallelism across a large number of workers. Federated learning (FL) goes beyond the data center setting and further considers decentralized data ownership, limited communication, and system and data heterogeneity across client nodes [17, 30, 36, 43]. In data-center training, communication efficiency is often addressed through compression techniques, such as quantization [4, 8], sparsification [44], and low-rank approximation (e.g.

PowerSGD) [46]. In FL, classical methods such as FedAvg and Local SGD reduce communication costs by performing multiple local updates between synchronization rounds [30, 43]. More advanced momentum-based variance-reduction schemes address node drift and stochastic noise under heterogeneous data distributions (e.g., SCAFFOLD [17, 18]). Another line of work explores communication compression techniques [3, 6, 14, 37, 44, 48]. Compared to FL, decentralized learning removes the central server entirely, relying instead on peer-to-peer gossip protocols to propagate information over a connected graph [32]. Classical approaches include D-PSGD [25], which averages models with neighbors, and CHOCO-SGD [21], which allows for compressed communication with error compensation. To address the steady-state error caused by data heterogeneity across nodes, Gradient Tracking methods [31, 35] have been proposed to align local updates with the global gradient direction.

We note, however, that these aforementioned methods are all developed for the conventional vector-based optimization paradigm, and do not explicitly account for nonlinear spectral transformations applied to matrix-valued momentum or gradient estimators.

3) Muon in Federated and Distributed Settings: Recently, several works have started to investigate Muon-style optimization in the FL setting. One line of work studied the theoretical benefits of Muon in FL, focusing on improved communication complexity, robustness to heavy-tailed noise, and learning-rate choices that are less sensitive to problem-dependent constants [2, 51]. Existing work also showed that naively using Muon as the local optimizer in FedAvg can fail to converge, and proposed a bias-corrected federated Muon scheme with control variates and inexact NS iterations [45]. Another federated Muon method investigated how to mitigate client drift in heterogeneous data settings by proposing a local-global alignment mechanism [26]. All these existing works showed that extending Muon to the federated learning setting is promising but non-trivial. For convenience in using these existing methods as baselines later, we refer to them as FedMuon-FA [51], FedMuon-SF [45], and FedMuon-LRSF [26] in this paper.

Our work is related to and yet fundamentally different from the above FedMuon-type methods in the following key aspects: Existing approaches primarily focused on properties of orthogonalized directions, bias correction caused by the non-commutativity of node averaging and local spectral mapping, and/or local-global alignment under heterogeneity. In contrast, our main goal is to build a *variance-reduced distributed SSBO backbone*. Specifically, we explicitly control the stochastic deviation of the matrix-valued momentum estimator *before* applying the nonlinear spectral mapping, and use this as a backbone to obtain a unified treatment of exact orthogonalization, finite-step NS approximation, and communication-efficient low-rank synchronization. Such variance-reduced mechanisms was not considered in the FedMuon-type baselines [26, 45, 51].

3 Problem Statement and Preliminaries

In this section, we will first state the problem formulation for distributed learning in the matrix-valued form in Section 3.1, which is followed by an overview on the basic idea of spectral mapping and exact orthogonalization in matrix-valued optimization in 3.2. Then, in Section 3.3, we will dive deeper to discuss several key challenges

in distributed SSBO-based algorithmic designs and theoretical analysis to further motivate this work.

3.1 Distributed Matrix-Valued Optimization

We consider a distributed optimization problem over matrix-valued parameters. Let $W \in \mathbb{R}^{m \times n}$ denote the model parameter, and suppose there are K nodes indexed by $k \in \{1, \dots, K\}$. Each node has access to a local data distribution $\mathcal{D}_k$ and defines a local objective $f^{(k)}(W) = \mathbb{E}_{\xi_k \sim \mathcal{D}_k} [f^{(k)}(W; \xi_k)]$, where $f^{(k)}(W; \xi)$ denotes the sample-wise loss. The global objective can be written as:

$$\min_{W \in \mathbb{R}^{m \times n}} f(W) := \frac{1}{K} \sum_{k=1}^K f^{(k)}(W).$$

In this work, we consider the nonconvex setting, where each f^k is nonconvex in W . The goal of solving the distributed optimization problem is to find an approximate stationary matrix-valued solution. In this paper, we use upper case letters to represent matrices. For a matrix $A \in \mathbb{R}^{m \times n}$, we let $\{\sigma_i(A)\}_{i=1}^{\min(m,n)}$ denote its singular values. We use $\|A\|_F$, $\|A\|_{\text{op}}$, and $\|A\|_*$ to denote the Frobenius norm, spectral norm, and nuclear norm, respectively: $\|A\|_F = (\sum_i \sigma_i(A)^2)^{1/2}$, $\|A\|_{\text{op}} = \max_i \sigma_i(A)$, and $\|A\|_* = \sum_i \sigma_i(A)$. These norms satisfy $\|A\|_{\text{op}} \leq \|A\|_F \leq \|A\|_*$. When A has rank at most r , we also have $\|A\|_* \leq \sqrt{r} \|A\|_F$.

3.2 Spectral Mapping and Orthogonalization for SSBO-Based Optimization

In matrix-valued optimization, the SSBO approach applies a nonlinear spectral mapping on a matrix-valued gradient or momentum estimator. In Muon-style methods, the spectral mapping is an exact orthogonalization. For a matrix $M \in \mathbb{R}^{m \times n}$ with singular value decomposition $M = U\Sigma V^\top$, we define $\text{Orth}(M) := UV^\top$. Muon-style optimizers use this spectral mapping on a momentum matrix M_t . In the centralized setting, an orthogonalized Muon update on the matrix weight M_t can be written as:

$$M_{t+1} = \beta M_t + (1 - \beta) \nabla f(W_t; \xi_t), \quad W_{t+1} = W_t - \eta_t \text{Orth}(M_{t+1}).$$

The key challenge in distributed SSBO is that $\text{Orth}(\cdot)$ is nonlinear, which results in several key challenges in distributed SSBO-based algorithmic design as stated next.

3.3 Key Challenges in Distributed SSBO-based Algorithmic Design

Although SSBO-based approaches have shown promises in centralized training, extending them to distributed settings suffers from several structural challenges that do not arise in conventional vector-based distributed optimization. These challenges stem from the same source: SSBO applies a nonlinear spectral mapping to a matrix-valued gradient or momentum estimator. In distributed settings, this nonlinearity interacts with various factors, such as data and system heterogeneity, and communication constraints. In what follows, we outline several key challenges to further motivate the design of DisMUSE in this work.

Challenge 1) Non-Commutativity and Aggregation Bias: The first consequence of the spectral mapping non-linearity is that

the spectral mapping does *not* commute with distributed aggregation. More precisely, let $M_t^{(k)}$ denote the local momentum estimator maintained by node k at iteration t , we have:

$$\frac{1}{K} \sum_{k=1}^K \text{Orth}(M_t^{(k)}) \neq \text{Orth}\left(\frac{1}{K} \sum_{k=1}^K M_t^{(k)}\right).$$

As a result, averaging locally orthogonalized momentum directions is *not* equivalent to the spectral direction of the averaged global momentum in the centralized setting. This differs fundamentally from vector-based distributed optimization, where commutativity holds due to the linear averaging operation.

This non-commutativity introduces systematic bias in distributed SSBO, especially when local objectives or stochastic gradients differ across nodes. Recent FedMuon-type analyses show that naively using local Muon-style updates within the FedAvg method [30] fails to reach a stationary solution [45]. The non-commutativity suggests a key design principle in DisMUSE: *the algorithm should synchronize and control the matrix-valued momentum estimators first before averaging locally transformed spectral directions.*

Challenge 2) Variance Amplification under Nonlinear Spectral Mapping: Even if aggregation is performed at the momentum level, distributed SSBO remains highly sensitive to stochastic noise. Specifically, let $M_t = M_t^{\text{det}} + E_t$, $\mathbb{E}[E_t] = 0$, where M_t^{det} is the deterministic momentum and E_t is a stochastic estimation error. In conventional vector-based updates, E_t emerges in the update in an additive fashion as follows:

$$W_{t+1} = W_t - \eta M_t = W_t - \eta M_t^{\text{det}} - \eta E_t.$$

In SSBO, however, the corresponding update perturbation becomes

$$\text{Orth}(M_t^{\text{det}} + E_t) - \text{Orth}(M_t^{\text{det}}),$$

which is not a linear or additive function of E_t . As a result, even if $\mathbb{E}[E_t] = 0$, one still has $\mathbb{E}[\text{Orth}(M_t)] \neq \text{Orth}(M_t^{\text{det}})$. Thus, stochastic noise in the momentum leads to a biased and directionally distorted spectral update estimator after orthogonalization. Therefore, in distributed SSBO, variance control is not merely preferred for efficiency improvement, but rather *necessary* for obtaining robust spectral updates. Our DisMUSE method addresses this challenge by employing a STORM-style recursive estimator [12] to reduce the variance of $\|E_t\|_F$ in the matrix-valued momentum before applying the spectral mapping.

Challenge 3): Orthogonalization Approximation Error: Our basic version of DisMUSE is based on exact orthogonalization. However, exact computation of $\text{Orth}(M)$ could be expensive for large matrix-valued parameters. In practical Muon-style implementations, the exact orthogonalization is typically approximated by a finite steps of low-complexity Newton-Schulz (NS) iterations. However, the NS approximation creates an additional layer of difficulty in theoretical analysis.

To see this, let $X_{t,q}$ denote the result after q NS iterations applied to M_t , then the actual update direction $X_{t,q}$ is *different* from $\text{Orth}(M_t)$ with the following approximation error: $X_{t,q} - \text{Orth}(M_t)$, which may not be negligible. What exacerbates the problem is the fact that this approximation error resides within the *nonlinear* spectral transformation, which could *not* be averaged out over time and may even propagate across iterations.. This motivates our NS

extension, where we theoretically quantify the required number of NS steps to allow the basic version of DisMUSE remain convergent with the same finite-time convergence rate.

Challenge 4) Communication Constraints: Lastly, distributed SSBO needs to take the communication cost of synchronizing matrix-valued momentum variables into consideration. For a matrix $M \in \mathbb{R}^{m \times n}$, a full communication requires transmitting mn scalars per synchronization, which can be prohibitive for large neural network models. This communication cost is particularly problematic in federated and bandwidth-constrained distributed environments.

A natural idea to address the communication challenge is to exploit the low-rank structure often observed in matrix-valued spectral updates. However, low-rank communication must be handled carefully, since low-rank factorization is yet another nonlinear operation that leads to non-commutativity, and naively averaging or discarding spectral information introduces further biases in the update. This challenge motivates our proposed low-rank approximation extension of DisMUSE, where the low-rank approximation error is managed judiciously.

Roadmap of Our DisMUSE Algorithmic Design: Addressing the aforementioned challenges motivates a three-stage development of DisMUSE. We begin by introducing a basic version of DisMUSE in Section 4, incorporating variance reduction, exact orthogonalization, and full-rank communication to mitigate aggregation bias and stochastic variance at the momentum level. We then extend this framework in Section 5 by adopting a finite-step NS orthogonalization approximation to enable a lower-complexity implementation. Finally, in Section 6, we further generalize DisMUSE to a low-rank communication variant and characterize the resulting accuracy-communication trade-off.

4 DisMUSE: Variance-reduced Distributed SSBO with Exact Orthogonalization

In this section, we start with the core algorithmic design of the basic version DisMUSE with exact orthogonalization (through exact SVD) and full-rank communication. The key idea is to synchronize the matrix-valued momentum estimator across nodes and to control its stochastic variance before applying the nonlinear spectral map.

1) Algorithm Description: We consider a distributed protocol with K nodes and communication period τ . Each node k maintains a local model $W_t^{(k)}$ and a matrix-valued momentum estimator $M_t^{(k)}$. In the t -th communication round, the nodes synchronize both model parameters and momentum estimators as follows: $W_t^{(k)} \leftarrow \frac{1}{K} \sum_{k'=1}^K W_t^{(k')}$, $M_t^{(k)} \leftarrow \frac{1}{K} \sum_{k'=1}^K M_t^{(k')}$. We note that synchronizing $M_t^{(k)}$ first is important because the spectral map is nonlinear, which implies non-commutativity: applying $\text{Orth}(\cdot)$ after momentum aggregation is different from averaging locally orthogonalized directions in general.

Between communication rounds, each node updates its momentum using a STORM-style recursive estimator [12]. Specifically, given a stochastic sample $\xi_t^{(k)}$, node k computes

$$M_t^{(k)} = \beta M_{t-1}^{(k)} + (1 - \beta) \nabla f^{(k)} \left(W_t^{(k)}; \xi_t^{(k)} \right) + \beta \left(\nabla f^{(k)} \left(W_t^{(k)}; \xi_t^{(k)} \right) - \nabla f^{(k)} \left(W_{t-1}^{(k)}; \xi_t^{(k)} \right) \right).$$

Algorithm 1 The basic version of DisMUSE with exact orthogonalization and full-rank communication.

```

1: Input: stepsize schedule  $\{\eta_t\}_{t \geq 0}$ , parameter  $\beta \in (0, 1)$ , communication period  $\tau \geq 1$ , mode  $\in \{\text{SVD}, \text{NS}\}$ , NS steps  $q$  (if mode=NS)
2: Initialize global model  $W_0$  and broadcast  $W_0$  to all nodes
3: Initialize  $W_0^{(k)} = W_0$  and  $M_0^{(k)} = \nabla f^{(k)}(W_0^{(k)}; \xi_0^{(k)})$  for each node  $k$ 
4: for  $t = 0, \dots, T-1$ , on each node  $k$  do
5:   if  $\text{mod}(t, \tau) = 0$  then
6:      $W_t^{(k)} \leftarrow \frac{1}{K} \sum_{k'=1}^K W_t^{(k')}$ ,
7:      $M_t^{(k)} \leftarrow \frac{1}{K} \sum_{k'=1}^K M_t^{(k')}$ 
8:   end if
9:   Sample mini-batch  $\xi_t^{(k)}$ 
10:   $M_t^{(k)} \leftarrow \beta M_{t-1}^{(k)} + (1-\beta) \nabla f(W_t^{(k)}; \xi_t^{(k)})$ 
11:     $+ \beta \left( \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}) \right)$ 
12:   $O_t^{(k)} \leftarrow \text{ORTHOGONALIZE}(M_t^{(k)}; \text{mode}, q)$ 
13:   $W_{t+1}^{(k)} \leftarrow W_t^{(k)} - \eta_t O_t^{(k)}$ 
14: end for

```

Algorithm 2 ORTHOGONALIZE(M ; mode, q).

```

1: if mode = SVD then
2:    $(U, S, V) \leftarrow \text{SVD}(M)$ 
3:   return  $UV^\top$ 
4: end if
5: if mode = NS then
6:    $\alpha \leftarrow \max\{1, \|M\|_F\}$ ,  $X_0 \leftarrow M/\alpha$ 
7:   for  $j = 1$  to  $q$  do
8:      $X_j \leftarrow p_\kappa(X_{j-1} X_{j-1}^\top) X_{j-1}$ 
9:   end for
10:  return  $X_q$ 
11: end if

```

The model is then updated by exact orthogonalization as follows:

$$W_{t+1}^{(k)} = W_t^{(k)} - \eta_t \text{Orth}(M_t^{(k)}).$$

The recursive correction term above reduces the stochastic deviation of $M_t^{(k)}$, which is essential before applying the nonlinear spectral mapping. The full algorithm is illustrated in Algorithm 1, with the SVD-based orthogonalization shown in Algorithm 2.

2) Convergence Analysis: Before conducting convergence analysis, we first introduce several useful assumptions that are standard in nonconvex distributed stochastic optimization.

ASSUMPTION 1 (FROBENIUS NORM LIPSCHITZ SMOOTHNESS). *Each local objective $f^{(k)}$ is L -smooth with respect to the Frobenius norm: for any $W, W' \in \mathbb{R}^{m \times n}$, $\|\nabla f^{(k)}(W) - \nabla f^{(k)}(W')\|_F \leq L\|W - W'\|_F$.*

This assumption generalizes the standard Lipschitz-smooth assumption from vector-valued to matrix-valued optimization problems [41]. To facilitate stochastic optimization convergence analysis, we use the following standard bounded variance assumption:

ASSUMPTION 2 (BOUNDED VARIANCE). *For each node k , the stochastic gradient is unbiased and has bounded variance: $\mathbb{E}[\nabla f^{(k)}(W; \xi_k)] = \nabla f^{(k)}(W)$, and $\mathbb{E}[\|\nabla f^{(k)}(W; \xi_k) - \nabla f^{(k)}(W)\|_F^2] \leq \sigma^2$.*

Further, we assume that the statistical data heterogeneity across nodes is bounded in the sense of gradient Frobenius norm:

ASSUMPTION 3 (BOUNDED HETEROGENEITY). *The local gradients satisfy $\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\nabla f^{(k)}(W) - \nabla f(W)\|_F^2] \leq \delta^2$, where $\delta \geq 0$ measures the degree of data heterogeneity.*

Note that when $\delta = 0$, the data distributions across nodes are homogeneous. In this work, we measure the convergence of our DisMUSE algorithm using a Frobenius-norm stationarity metric, which generalizes the conventional Euclidean stationarity criterion to matrix-valued problems.

Definition 4.1 (ϵ -Frobenius-norm Stationarity). A solution $\hat{W}$ is an ϵ -Frobenius stationary solution of $f(\cdot)$ if $\|\nabla f(\hat{W})\|_F \leq \epsilon$.

Next, we present the main convergence result of the basic DisMUSE, which is based on exact orthogonalization and full-rank communications. The analysis shows that once the stochastic variance of the matrix-valued momentum is well controlled before spectral mapping, our DisMUSE method enjoys the *same* finite-time convergence rate as variance-reduced nonconvex distributed optimization. Let $\bar{W}_t := \frac{1}{K} \sum_{k=1}^K W_t^{(k)}$ represent the averaged model. We have:

THEOREM 4.2 (CONVERGENCE OF DISMUSE WITH EXACT ORTHOGONALIZATION AND FULL-RANK COMMUNICATION). *Under Assumption 1, 2 and 3 hold, by choosing $\eta_t = \eta$, $\forall t$, the averaged gradient norm generated by Algorithm 1 satisfies, for any $T \geq 1$,*

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F \\ & \leq \frac{\mathbb{E} f(\bar{W}_0) - f_*}{\eta T} + 16\eta\tau^2 Lr + 8\sqrt{r}\tau\sigma + 4\sqrt{r}\tau\delta + 4\eta\tau Lr + \frac{\eta r L}{2} \\ & \quad + 2\sqrt{r} \sqrt{\frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{2L^2\eta^2 r}{1-\beta}}. \end{aligned}$$

In particular, choosing $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1-\beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, and assuming $T > K$, we obtain

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F = O\left(\frac{1}{K^{1/3}T^{1/3}}\right).$$

The result in Theorem 4.2 says that incorporating recursive variance control in DisMUSE yields an $O((KT)^{-1/3})$ stationarity guarantee under exact orthogonalization. Compared with the $O((KT)^{-1/4})$ rate under comparable assumptions in existing federated SSBO methods based on standard momentum (FedMuon-FA [51], FedMuon-SF [45] and FedMuon-LRSF [26]), our convergence result demonstrates that actively reducing the variance in stochastic gradient noise leads to provably a faster convergence. This highlights the importance of variance-reduced estimators when applying nonlinear spectral mappings in distributed settings, where update directions are particularly sensitive to gradient estimation error.

5 DisMUSE with Newton-Schulz Orthogonalization Approximation

In the previous section, we have shown the strong convergence performance of the basic DisMUSE method based on exact orthogonalization. However, computing exact $\text{Orth}(M)$ through SVD-based

orthogonalization can be expensive for large matrices. In practical Muon-style implementations, exact orthogonalization is often replaced by a finite number of low-complexity NS steps. In this section, we will establish the theoretical convergence guarantees of our DisMUSE method with NS orthogonalization approximation.

1) Preliminaries on the Newton-Schulz Approximation and Algorithmic Updates in DisMUSE: NS approximates the exact orthogonal factor $\text{Orth}(M_t)$, which is equal to $M_t(M_t^\top M_t)^{-1/2}$. Starting from a rescaled matrix of M_t denoted as $X_{t,0} = M_t/\alpha_t$, with $\alpha_t \geq \max\{1, \|M_t\|_F\}$ to ensure $\|X_{t,0}\|_F \leq 1$, NS approximation can be iteratively written as:

$$X_{t,j+1} = p_\kappa(X_{t,j}X_{t,j}^\top)X_{t,j}, \quad j = 0, \dots, q-1,$$

where p_κ is a polynomial approximation of $\lambda^{-1/2}$ defined as $p_\kappa(\lambda) = \sum_{s=0}^\kappa c_s(1-\lambda)^s$, $c_s = \frac{(2s)!}{4^s(s!)^2} > 0$. Intuitively, this iteration pushes the singular values of $X_{t,j}$ toward 1, so $X_{t,q}$ approaches the exact orthogonal factor $\text{Orth}(M_t)$.

Due to space limitations and to avoid repetition, we omit the description of DisMUSE with the NS approximation, as it differs from the basic DisMUSE only in the orthogonalization step. In what follows, we outline how to replace the exact orthogonalization with a finite-step NS approximation: In t -th iteration at node k , we first apply q NS steps on $M_t^{(k)}$ to obtain $X_{t,q}^{(k)}$, which serves as an approximation to the exact orthogonal factor. Then, the update direction is computed as:

$$O_t^{(k)} = X_{t,q}^{(k)}. \quad (1)$$

2) Convergence Analysis: To conduct convergence analysis of DisMUSE under NS orthogonalization approximation, we adopt the following assumption:

ASSUMPTION 4 (SPECTRAL NORM LIPSCHITZ SMOOTHNESS). Each local objective $f^{(k)}$ is L_* -smooth with respect to the spectral/nuclear norm pair: for any $W, W' \in \mathbb{R}^{m \times n}$, it holds that $\|\nabla f^{(k)}(W) - \nabla f^{(k)}(W')\|_* \leq L_* \|W - W'\|_{\text{op}}$.

This assumption captures smoothness properties aligned with the spectral structure of matrix-valued gradients and will be used when measuring stationarity in terms of the nuclear norm [11]. Moreover, for more convenience in analyzing NS steps in spectral approximation, we adopt the following nuclear-norm stationarity.

Definition 5.1 (Nuclear-norm stationarity). A solution $\widehat{W}$ is said to be an ϵ -nuclear stationary solution of $f(\cdot)$ if $\|\nabla f(\widehat{W})\|_* \leq \epsilon$.

Note that, since $\|A\|_F \leq \|A\|_*$, nuclear-norm stationarity implies Frobenius-norm stationarity (i.e., our convergence result is stronger than needed for the Frobenius norm), which reconciles with our previous convergence results for the basic DisMUSE method.

To quantify the NS approximation error after q steps, let $X_{t,q}$ denote the resulting matrix at iteration t and let Π_t be the orthogonal projector onto the range of the corresponding momentum matrix M_t . We define orthogonality residual as $\delta_{t,q} := \|\Pi_t - X_{t,q}X_{t,q}^\top\|_{\text{op}}$ and polar approximation error as $\varepsilon_{t,q} := \|X_{t,q} - \text{Orth}(M_t)\|_{\text{op}}$. These quantities capture the deviation from perfect orthogonality and the error relative to exact orthogonalization, respectively. Then, we invoke a known quantitative bound on the approximation error induced by the NS iteration, which characterizes how the number

of iterations q controls the deviation from exact orthogonalization and will be used in our convergence analysis.

LEMMA 5.2 (RESTATED FROM [19]). For the Newton-Schulz polynomial with degree κ and any t , $\delta_{t,q} \leq \delta_{t,0}^{(\kappa+1)^q}$. Consequently, the polar approximation error and the associated amplification factor satisfy $\varepsilon_q \leq 1 - \sqrt{1 - \delta_0^{(\kappa+1)^q}} \leq \delta_0^{(\kappa+1)^q}$, and $\chi_q := \frac{1}{1 - \varepsilon_q} \leq \frac{1}{\sqrt{1 - \delta_0^{(\kappa+1)^q}}}$, where $\delta_0 := \sup_t \delta_{t,0} < 1$.

This result implies that the approximation error ε_q decays at a doubly exponential rate in the number of NS steps q , and the associated factor χ_q converges to 1 at the same rate. Our main convergence result for the DisMUSE method with NS orthogonalization approximation is stated as follows:

THEOREM 5.3. Let $\chi_q = \frac{1}{1 - \varepsilon_q} > 0$ be a constant with ε_q being defined as in Lemma 5.2. Under Assumption 4,2 and 3, by choosing $\eta_t = \eta$, $\forall t$, the averaged gradient generated by Algorithm 1 with NS orthogonalization approximation satisfies, for any $T \geq 1$,

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_* \\ & \leq \chi_q \left(\frac{\mathbb{E} f(\bar{W}_0) - f_*}{\eta T} + 16\eta\tau^2 L (1 + \varepsilon_q) + 8\sqrt{r}\tau\sigma \right. \\ & \quad \left. + 4\sqrt{r}\tau\delta + 4\eta\tau L (1 + \varepsilon_q) + \frac{\eta(1 + \varepsilon_q)^2 L}{2} \right. \\ & \quad \left. + 2\sqrt{\frac{r\sigma^2}{KB_0(1 - \beta)T} + \frac{2r(1 - \beta)\sigma^2}{K} + \frac{2L^2\eta^2(1 + \varepsilon_q)^2}{1 - \beta}} \right). \end{aligned}$$

Let $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1 - \beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, if $T > K$, then we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_* = \chi_q O\left(\frac{1}{K^{1/3}T^{1/3}}\right).$$

Two important remarks of Theorem 5.3 are in order. First, our convergence bound involves a multiplicative factor χ_q due to the finite-step NS orthogonalization. By Lemma 5.2, χ_q converges rapidly to 1 as q increases. Thus, DisMUSE with NS asymptotically approaches the same convergence rate as the exact orthogonalization while avoiding its computational cost.

Second, compared to FedMuon-SF [45], whose analysis only establishes convergence for an arbitrary number of NS iterations under fixed parameters, our convergence result is stronger in two aspects: 1) We provide a quantitative characterization of the discrepancy between the exact orthogonalization and its finite-step NS approximation, rather than only proving convergence in a qualitative sense; and 2) We show that our method still preserves the $O((KT)^{-1/3})$ convergence rate, which is stronger than the rate established in FedMuon-SF. Meanwhile, the other two FedMuon works [26, 51] only analyze the exact orthogonalization version and therefore do not capture the effect of practical inexact orthogonalization.

Algorithm 3 DisMUSE with Low-Rank Communication

```

1: for  $t = 0, \dots, T - 1$  do
2:   for each node  $k$  in parallel do
3:     Local spectral step:
4:      $O_t^{(k)} \leftarrow \text{Orth}(M_t^{(k)}) \{ \text{SVD or Newton-Schulz} \}$ 
5:     Local model update:  $W_{t+1}^{(k)} \leftarrow W_t^{(k)} - \eta O_t^{(k)}$ 
6:     Sample  $\xi_t^{(k)}$ 
7:     Local STORM update:
8:      $M_{t+1}^{(k)} \leftarrow \beta M_t^{(k)} + (1 - \beta) \nabla f^{(k)}(W_{t+1}^{(k)}; \xi_t^{(k)})$ 
9:      $+ \beta (\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}))$ 
10:    end for
11:    if  $(t + 1) \bmod \tau = 0$  then
12:      Upload:
13:      nodes send  $W_{t+1}^{(k)}$  and  $(U_r^{(k)}, \Sigma_r^{(k)}, V_r^{(k)}) = \text{SVD}_r(M_{t+1}^{(k)})$ 
14:      Aggregate:
15:       $W_{t+1} \leftarrow \frac{1}{K} \sum_k W_{t+1}^{(k)}$ ,  $M_{t+1,r} \leftarrow \frac{1}{K} \sum_k U_r^{(k)} \Sigma_r^{(k)} (V_r^{(k)})^\top$ 
16:      Re-compress & broadcast:
17:       $(U_r, \Sigma_r, V_r) = \text{SVD}_r(M_{t+1,r})$ ; broadcast  $(W_{t+1}, U_r, \Sigma_r, V_r)$ 
18:      Sync:  $W_{t+1}^{(k)} \leftarrow W_{t+1}$ ,  $M_{t+1}^{(k)} \leftarrow U_r \Sigma_r V_r^\top$ 
19:    end if
20:  end for

```

6 DisMUSE with Low-Rank Communications

In this section, we focus on a communication-efficient variant of DisMUSE, which is termed as DisMUSE-LR and designed for bandwidth-constrained distributed learning. The goal is to reduce communication cost while preserving the convergence guarantees of the basic DisMUSE algorithm.

1) Algorithm Description: For the first time step ($t = 0$), we initialize as $M_0 = \frac{1}{B_0} \sum_{i=1}^{B_0} \nabla f(W_0; \xi_0^i)$, where $B_0 = O(T^{1/3})$ is the batch size used in the initial iteration. In each communication round, the nodes do not transmit full matrix-valued updates. Instead, each node k computes a rank- r truncated SVD of its local momentum matrix $M_{t+1,r}^{(k)} \approx U_r^{(k)} \Sigma_r^{(k)} (V_r^{(k)})^\top$ and uploads the factorized representation $(U_r^{(k)}, \Sigma_r^{(k)}, V_r^{(k)})$ to the server. Meanwhile, nodes upload their local models $W_{t+1}^{(k)}$.

Upon receiving the factorized updates, the server first reconstructs the corresponding matrices and performs linear aggregation as follows: $M_{t+1,r} = \frac{1}{K} \sum_{k=1}^K U_r^{(k)} \Sigma_r^{(k)} (V_r^{(k)})^\top$. Performing this reconstruction step first is essential, as low-rank factorization is again *not* a linear operation and breaks commutativity. The server then applies a rank- r truncated SVD to the aggregated matrix, $M_{t+1,r} \approx U_r \Sigma_r V_r^\top$, and broadcasts the factorized representation (U_r, Σ_r, V_r) back to all nodes. Also, the server computes the averaged model $W_{t+1} = \frac{1}{K} \sum_k W_{t+1}^{(k)}$ and broadcasts it to the nodes.

Upon receiving the information from the server, each node reconstructs the synchronized momentum matrix locally as $M_{t+1,r}^{(k)} \leftarrow U_r \Sigma_r V_r^\top$, and continues local updates until the next communication round. To conclude the algorithmic description, we summarize the DisMUSE-LR algorithm in Algorithm 3.

2) Convergence Analysis under Low-Rank Communications: Note that low-rank communications introduce a truncation error by discarding singular components beyond rank r . To

quantify this impact, we define a truncation residual as follows: $\rho_t := \max_{k \in [K]} (\sum_{i=r+1}^d (\sigma_i^{(k)})^2)^{1/2} = \max_{k \in [K]} \|M_t^{(k)} - M_{t,r}^{(k)}\|_F$, where $\{\sigma_i^{(k)}\}$ are the singular values of $M_t^{(k)}$ and $d = \min\{m, n\}$. Let $\rho_r := \sup_{t \geq 0} \rho_t$. The quantity ρ_r captures the intrinsic bias introduced by low-rank communication and depends only on the spectral tail of the update matrices. We now state the main convergence result for DisMUSE-LR.

THEOREM 6.1. *Under Assumption 1,2 and 3, by choosing $\eta_t = \eta$, $\forall t$, the averaged gradient norm generated by Algorithm 3 satisfies, for any $T \geq 1$,*

$$\begin{aligned} & \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F \\ & \leq \frac{\mathbb{E} f(\bar{W}_0) - f_*}{\eta T} + 16\eta \tau^2 L r + 8\sqrt{r} \tau \sigma + 4\sqrt{r} \tau \delta + \eta(6\sqrt{r} + \sqrt{d-r}) \rho_r \\ & \quad + 4\eta \tau L r + \frac{\eta r L}{2} + 2\sqrt{r} \sqrt{\frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{2L^2\eta^2 r}{1-\beta}}. \end{aligned}$$

Let $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1 - \beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, if $T > K$, then we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F = O\left(\frac{1}{K^{1/3}T^{1/3}}\right) + \frac{(6\sqrt{r} + \sqrt{d-r}) \rho_r}{K^{1/3}T^{1/3}}.$$

We note that the bound in Theorem 6.1 consists of the same $O((KT)^{-1/3})$ term as in the full-rank DisMUSE algorithm and an additive bias term proportional to the spectral tail ρ_r , which captures the effect of low-rank truncation. Four important remarks for Theorem 6.1 are in order:

Remark 1 (Impact of Low-rank Communications). We note that the truncation error ρ_r reflects the spectral tail of the update matrices. If $M_t^{(k)}$ is exactly rank r , then $\rho_r = 0$ and the low-rank method matches the full-rank guarantee. If the singular values decay, ρ_r decreases with r , yielding a controllable accuracy-communication trade-off: reducing communication costs by decreasing r increases the truncation bias, while larger r improves accuracy at the expense of higher communication costs. Without spectral decay, ρ_r remains non-negligible and appears as an additive bias term in the convergence bound.

Remark 2 (Communication Costs Reduction). We now quantify the communication cost of the proposed low-rank protocol and compare it with standard dense transmission. For a momentum matrix $M \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the original dense full-rank transmissions require $d_{\text{out}} d_{\text{in}}$ scalars. In contrast, the rank- r low-rank communications transmit factors (U_r, Σ_r, V_r) , reducing the cost to $r(d_{\text{out}} + d_{\text{in}} + 1)$. This yields a compression ratio of approximately:

$$\frac{d_{\text{out}} d_{\text{in}}}{r(d_{\text{out}} + d_{\text{in}})},$$

providing substantial bandwidth savings when $r \ll \min\{d_{\text{out}}, d_{\text{in}}\}$. When $d_{\text{out}} = d_{\text{in}} = d$, the compression ratio scales as $\frac{d^2}{r(2d+1)} \approx \frac{d}{2r}$. For a typical Transformer layer in LLaMA-7B with $d = 4096$, choosing $r = 32$ reduces the communication payload by approximately $64\times$ (from about 16MB to 0.25MB per matrix). This significant reduction makes DisMUSE-LR updates compatible with bandwidth-constrained distributed settings.

Remark 3 (Relation to NS Approximation). Similar to the NS approximation used in DisMUSE, DisMUSE-LR can also replace the local rank r truncated SVD step between communication rounds with a finite number of NS iterations. The corresponding approximation error is characterized in Section 5, where we show that finite-step NS orthogonalization preserves the asymptotic convergence order while significantly reducing the computational cost.

Remark 4. Our analysis directly tracks the low-rank approximation error and establishes an explicit convergence trade-off between low-rank communications and the bias introduced by rank- r truncated SVD. In contrast, although the FedMuon-LRSF method in [26] also considers low-rank approximation with rank r for the orthogonalized update gradients, its convergence guarantee is derived based on the full-rank update, and thus does not explicitly capture the error induced by rank- r truncation.

7 Experiments

In this section, we conduct experiments to evaluate DisMUSE and validate the theoretical findings of our proposed DisMUSE algorithms.

7.1 CIFAR-10 Classification with ResNet-18

1) Dataset and Model: We evaluate our algorithms on the CIFAR-10 using ResNet-18 datasets. To avoid the instability of Batch Normalization under small local batches in federated training, we replace all BN layers with Group Normalization (two groups).

2) Network Setting: We consider $K = 5$ nodes with full participation in each round. The training data are partitioned across nodes using LDA with concentration parameter $\alpha = 0.5$, which induces moderate data heterogeneity.

3) Experiment Parameters: We use local batch size $B = 32$ and $E = 5$ local epochs per round. For approximate orthogonalization, we use $q = 5$ NS iterations. For communication compression, the truncation rank is set to $r = 32$ for convolutional kernels.

4) Baselines: We compare DisMUSE with FedAvg [30], FedAdam [36], FedMuon-FA [51] and FedMuon-SF [45]. DisMUSE Hybrid denotes the low-rank DisMUSE-LR that uses NS iteration for local orthogonalization approximation.

5) Experimental Results: It can be seen that DisMUSE with NS outperforms FedAdam and the other baselines across the tested settings. In addition, DisMUSE with exact SVD achieves better performance than FedMuon-FA, which does not use variance reduction (VR), which clearly shows the benefit of the VR design in our algorithms. We also observe that the NS-based variant of DisMUSE is faster than the basic DisMUSE with exact SVD orthogonalization, which verifies the benefits of using NS as an efficient orthogonalization approximation scheme. Lastly, the strong performance under different number of nodes K , different NS steps q , and different low ranks r further confirms the effectiveness and robustness of our proposed algorithms.

7.2 FashionMNIST Classification with a Two-layer CNN

1) Dataset and Model: We evaluate our algorithm on Fashion-MNIST using a two-layer CNN. The goal of this experiment is to compare the behavior of three variants of DisMUSE.

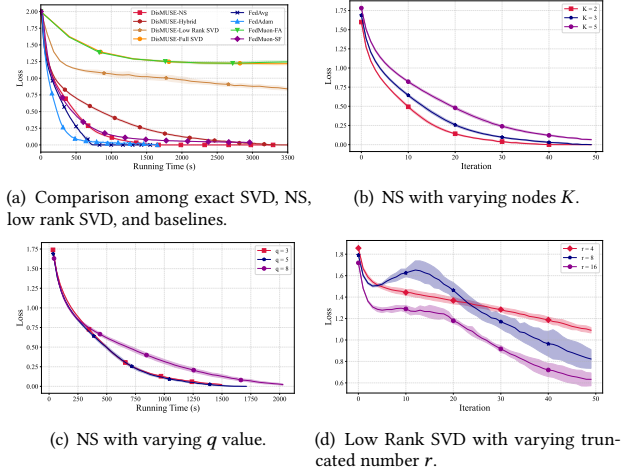


Figure 1: Test Loss on CIFAR-10 with ResNet-18.

2) Network Setting: We consider $K = 20$ nodes with full participation in each round. The training data are partitioned across nodes using LDA with concentration parameter $\alpha = 0.5$.

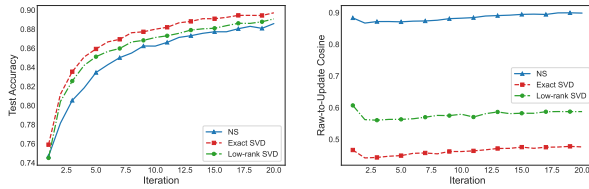
3) Implementation Details: We use a CNN with two 3×3 convolutional layers (32 and 64 channels), where each layer is followed by ReLU and 2×2 max pooling, together with a two-layer MLP head with a hidden dimension of 128 and an output dimension of 10. Unless otherwise specified, each client is assigned 3000 local training examples, uses a local batch size of $B = 128$. For orthogonalization approximation, we employ $q = 5$ NS iterations. For the low-rank communications, we set the truncation rank to $r = 16$. We use a 0.02 learning rate for DisMUSE and alg with low-rank communications, while the NS variant uses a 0.10 learning rate.

4) Comparisons: We compare the basic DisMUSE with exact orthogonalization and full-rank communication, DisMUSE with NS approximation, and DisMUSE-LR with low-rank communications.

5) Experimental Results: The results show clear differences among the three variants of DisMUSE. The NS-based variant achieves performance close to the exact SVD orthogonalization variant, while being computationally more efficient, which supports its benefits as an approximation to exact orthogonalization in practice. The DisMUSE-LR variant remains effective and provides an additional trade-off between performance and efficiency. In addition to training performance, we also study the directional effect of orthogonalization using cosine similarity. A cosine value closer to 1 indicates stronger directional alignment, while a value closer to 0 indicates a larger directional change. Specifically, we consider the *Raw-to-Update Cosine* between the raw and orthogonalized updates. Each matrix parameter is flattened into a vector before the computation. These cosine-similarity-based analysis further reveals the effect of orthogonalization: the NS-based update stays more aligned with the original update direction, whereas the SVD-based variants perform a more aggressive directional perturbation.

8 Conclusion

Singular-spectrum-based optimizers (SSBO) enable geometry-aware updates but have been largely limited to centralized training so



(a) Comparison among exact SVD, NS, (b) Cosine Similarity to the Original Up-date Direction.

Figure 2: Comparison of orthogonalization variants in performance and directional alignment.

far. In this paper, we proposed DisMUSE, a distributed singular-spectrum-based optimization framework that integrates Muon-style orthogonalized momentum with recursive variance reduction and applies to both data-center and federated settings. Our analysis first established an $O((KT)^{-1/3})$ convergence rate for the basic variance-reduced DisMUSE method. We then showed that finite-step NS orthogonalization preserves the same convergence rate up to a constant factor. Lastly, we characterized the accuracy-communication trade-off of our DisMUSE with low-rank communications. These results provide a first step toward principled and communication-efficient SSBO in distributed environments.

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Table 1: Main parameters used in DisMUSE.

Parameter	Definition
K	Number of distributed nodes.
T	Total number of optimization iterations.
η	Constant model-update stepsize.
β	STORM momentum parameter, where $\beta \in (0, 1)$.
τ	Communication period; synchronization occurs every τ local steps.
B_0	Initial mini-batch size used to initialize the momentum estimator.
r	Rank bound of the update direction; also the truncation rank in DisMUSE-LR.
q	Number of Newton–Schulz iterations.
L	Frobenius-norm smoothness constant.
L_*	Spectral/nuclear norm-pair smoothness constant.
σ^2	Upper bound on the stochastic-gradient variance.
δ^2	Upper bound on gradient heterogeneity across nodes.
ε_q	Uniform error between the q -step NS direction and exact orthogonalization.
χ_q	NS approximation factor, defined as $\chi_q = (1 - \varepsilon_q)^{-1}$.
ρ_r	Uniform rank- r truncation residual in DisMUSE-LR.

Proof roadmap. For exact orthogonalization, Lemma A.1 establishes the one-step descent inequality. Lemmas A.2, A.3, and A.4 bound the model-consensus error, momentum-consensus error, and averaged STORM estimation error, respectively. Substituting these bounds into the descent inequality and telescoping over t gives Theorem 4.2.

For finite-step Newton–Schulz orthogonalization, Lemma 5.2 relates the number of NS steps q to the approximation error ε_q and the factor χ_q . Lemmas B.1–B.4 repeat the same descent and error decomposition with the additional NS factors, yielding Theorem 5.3.

For low-rank communication, the proof inserts and subtracts the rank- r truncated momentum matrices. All additional truncation terms are bounded by ρ_r , which leads to the extra low-rank bias term and gives Theorem 6.1.

A Proof of theorem 4.2

LEMMA A.1. *Suppose Assumption 1,4,2 and 3 hold. Let $f(W) = \frac{1}{K} \sum_{k=1}^K f^{(k)}(W)$, $\nabla f(W) = \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W)$, then the following descent inequality holds:*

$$\begin{aligned}
f(\bar{W}_{t+1}) \leq & f(\bar{W}_t) - \eta \|\nabla f(\bar{W}_t)\|_F + 2\eta\sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F + 2\eta\sqrt{r} \frac{L}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F + 2\eta\sqrt{r} \\
& + 2\eta\sqrt{r} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F
\end{aligned}$$

PROOF. By L -smoothness of f , we have

$$f(\bar{W}_{t+1}) \leq f(\bar{W}_t) + \langle \nabla f(\bar{W}_t), \bar{W}_{t+1} - \bar{W}_t \rangle + \frac{L}{2} \|\bar{W}_{t+1} - \bar{W}_t\|_F^2. \quad (2)$$

The update rule implies

$$\bar{W}_{t+1} - \bar{W}_t = -\eta \frac{1}{K} \sum_{k=1}^K U_t^{(k)} (V_t^{(k)})^\top,$$

and hence, substituting into (2),

$$f(\bar{W}_{t+1}) \leq f(\bar{W}_t) - \eta \left\langle \nabla f(\bar{W}_t), \frac{1}{K} \sum_{k=1}^K U_t^{(k)} (V_t^{(k)})^\top \right\rangle + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K U_t^{(k)} (V_t^{(k)})^\top \right\|_F^2. \quad (3)$$

We expand the inner product by inserting and subtracting $\bar{M}_t$ and $M_t^{(k)}$:

$$\begin{aligned}
& -\eta \left\langle \nabla f(\bar{W}_t), U_t^{(k)} (V_t^{(k)})^\top \right\rangle \\
& = -\eta \left\langle \nabla f(\bar{W}_t) - \bar{M}_t, U_t^{(k)} (V_t^{(k)})^\top \right\rangle - \eta \left\langle \bar{M}_t - M_t^{(k)}, U_t^{(k)} (V_t^{(k)})^\top \right\rangle - \eta \left\langle M_t^{(k)}, U_t^{(k)} (V_t^{(k)})^\top \right\rangle.
\end{aligned}$$

Using Cauchy-Schwarz and $\|U_t^{(k)}(V_t^{(k)})^\top\|_F \leq \sqrt{r}$, we obtain

$$\begin{aligned} -\langle \nabla f(\bar{W}_t) - \bar{M}_t, U_t^{(k)}(V_t^{(k)})^\top \rangle &\leq \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F \|U_t^{(k)}(V_t^{(k)})^\top\|_F \leq \sqrt{r} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F, \\ -\langle \bar{M}_t - M_t^{(k)}, U_t^{(k)}(V_t^{(k)})^\top \rangle &\leq \|\bar{M}_t - M_t^{(k)}\|_F \|U_t^{(k)}(V_t^{(k)})^\top\|_F \leq \sqrt{r} \|\bar{M}_t - M_t^{(k)}\|_F. \end{aligned}$$

By the choice of $U_t^{(k)}(V_t^{(k)})^\top$,

$$\langle M_t^{(k)}, U_t^{(k)}(V_t^{(k)})^\top \rangle = \|M_t^{(k)}\|_*.$$

Moreover,

$$\left\| \frac{1}{K} \sum_{k=1}^K U_t^{(k)}(V_t^{(k)})^\top \right\|_F^2 \leq \frac{1}{K} \sum_{k=1}^K \|U_t^{(k)}(V_t^{(k)})^\top\|_F^2 \leq r.$$

Substituting these bounds into (3) gives

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* + \eta \sqrt{r} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F \\ &\quad + \eta \sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F + \frac{\eta^2 r L}{2}. \end{aligned} \quad (4)$$

Next, we relate $\sum_k \|M_t^{(k)}\|_*$ to $\|\bar{M}_t\|_*$. By the triangle inequality and $\|A\|_* \leq \sqrt{r}\|A\|_F$,

$$\|\bar{M}_t\|_* \leq \|\bar{M}_t - M_t^{(k)}\|_* + \|M_t^{(k)}\|_* \leq \sqrt{r} \|\bar{M}_t - M_t^{(k)}\|_F + \|M_t^{(k)}\|_*,$$

which yields

$$\|M_t^{(k)}\|_* \geq \|\bar{M}_t\|_* - \sqrt{r} \|\bar{M}_t - M_t^{(k)}\|_F.$$

Averaging over k gives

$$-\frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* \leq -\|\bar{M}_t\|_* + \sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F.$$

Substituting this into (4) yields

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta \|\bar{M}_t\|_* + \eta \sqrt{r} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F \\ &\quad + 2\eta \sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F + \frac{\eta^2 r L}{2}. \end{aligned} \quad (5)$$

We now compare $\|\bar{M}_t\|_*$ and $\|\nabla f(\bar{W}_t)\|_F$. Using $\|A\|_F \leq \|A\|_* \leq \sqrt{r}\|A\|_F$ and the triangle inequality,

$$\begin{aligned} \|\nabla f(\bar{W}_t)\|_F &\leq \|\nabla f(\bar{W}_t)\|_* = \|\nabla f(\bar{W}_t) - \bar{M}_t + \bar{M}_t\|_* \\ &\leq \|\nabla f(\bar{W}_t) - \bar{M}_t\|_* + \|\bar{M}_t\|_* \\ &\leq \sqrt{r} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F + \|\bar{M}_t\|_*. \end{aligned}$$

Rearranging gives

$$-\|\bar{M}_t\|_* \leq -\|\nabla f(\bar{W}_t)\|_F + \sqrt{r} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F.$$

Substituting into (5) yields

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta \|\nabla f(\bar{W}_t)\|_F + 2\eta \sqrt{r} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F \\ &\quad + 2\eta \sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F + \frac{\eta^2 r L}{2}. \end{aligned} \quad (6)$$

To decompose $\|\nabla f(\bar{W}_t) - \bar{M}_t\|_F$, we add and subtract the local true gradients:

$$\begin{aligned} \|\nabla f(\bar{W}_t) - \bar{M}_t\|_F &= \left\| \nabla f(\bar{W}_t) - \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) + \frac{1}{K} \sum_{k=1}^K (\nabla f^{(k)}(W_t^{(k)}) - M_t^{(k)}) \right\|_F \\ &\leq \left\| \nabla f(\bar{W}_t) - \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) \right\|_F + \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F. \end{aligned}$$

Since $f^{(k)}$ is L -smooth,

$$\|\nabla f^{(k)}(\bar{W}_t) - \nabla f^{(k)}(W_t^{(k)})\|_F \leq L \|\bar{W}_t - W_t^{(k)}\|_F,$$

and using

$$\nabla f(\bar{W}_t) = \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(\bar{W}_t),$$

we obtain

$$\begin{aligned} \left\| \nabla f(\bar{W}_t) - \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) \right\|_F &= \left\| \frac{1}{K} \sum_{k=1}^K (\nabla f^{(k)}(\bar{W}_t) - \nabla f^{(k)}(W_t^{(k)})) \right\|_F \\ &\leq \frac{1}{K} \sum_{k=1}^K \|\nabla f^{(k)}(\bar{W}_t) - \nabla f^{(k)}(W_t^{(k)})\|_F \\ &\leq \frac{L}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F. \end{aligned}$$

Combining the above, we arrive at

$$\|\nabla f(\bar{W}_t) - \bar{M}_t\|_F \leq \frac{L}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F + \frac{1}{K} \sum_{k=1}^K \|\nabla f^{(k)}(W_t^{(k)}) - M_t^{(k)}\|_F.$$

Substituting this bound into (6) yields

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta \|\nabla f(\bar{W}_t)\|_F + 2\eta\sqrt{r} \frac{L}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F \\ &\quad + 2\eta\sqrt{r} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F + 2\eta\sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F \\ &\quad + \frac{\eta^2 r L}{2}. \end{aligned}$$

This completes the proof. $\square$

LEMMA A.2. *Suppose Assumption 1,2 and 3 hold. At the beginning of each communication period of length τ , the models satisfy the synchronization condition $W_{s\tau}^{(k)} = \bar{W}_{s\tau}, \forall k, \forall s \in \mathbb{N}$. Then for any $t \geq 0$, we have*

$$\frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F \leq 2\eta\tau\sqrt{r}.$$

PROOF. Let $s_t = \lfloor t/\tau \rfloor$. Unrolling the updates from the beginning of the current communication round,

$$W_t^{(k)} = W_{s_t\tau}^{(k)} - \eta \sum_{t'=s_t\tau}^{t-1} U_{t'}^{(k)} (V_{t'}^{(k)})^\top, \quad \bar{W}_t = \bar{W}_{s_t\tau} - \eta \sum_{t'=s_t\tau}^{t-1} \frac{1}{K} \sum_{k'=1}^K U_{t'}^{(k')} (V_{t'}^{(k')})^\top.$$

By the synchronization condition $W_{s_t\tau}^{(k)} = \bar{W}_{s_t\tau}$, we obtain

$$\bar{W}_t - W_t^{(k)} = -\eta \sum_{t'=s_t\tau}^{t-1} \left(\frac{1}{K} \sum_{k'=1}^K U_{t'}^{(k')} (V_{t'}^{(k')})^\top - U_{t'}^{(k)} (V_{t'}^{(k)})^\top \right).$$

Thus,

$$\begin{aligned} \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F &= \eta \frac{1}{K} \sum_{k=1}^K \left\| \sum_{t'=s_t\tau}^{t-1} \left(U_{t'}^{(k)} (V_{t'}^{(k)})^\top - \frac{1}{K} \sum_{k'=1}^K U_{t'}^{(k')} (V_{t'}^{(k')})^\top \right) \right\|_F \\ &\leq \eta \frac{1}{K} \sum_{k=1}^K \left\| \sum_{t'=s_t\tau}^{t-1} U_{t'}^{(k)} (V_{t'}^{(k)})^\top \right\|_F \\ &\quad + \eta \frac{1}{K} \sum_{k=1}^K \left\| \sum_{t'=s_t\tau}^{t-1} \frac{1}{K} \sum_{k'=1}^K U_{t'}^{(k')} (V_{t'}^{(k')})^\top \right\|_F. \end{aligned} \tag{7}$$

For the first term, using the triangle inequality and $\|U_{t'}^{(k)} (V_{t'}^{(k)})^\top\|_F \leq \sqrt{r}$,

$$\left\| \sum_{t'=s_t\tau}^{t-1} U_{t'}^{(k)} (V_{t'}^{(k)})^\top \right\|_F \leq \sum_{t'=s_t\tau}^{t-1} \|U_{t'}^{(k)} (V_{t'}^{(k)})^\top\|_F \leq (t - s_t\tau) \sqrt{r} \leq \tau\sqrt{r}.$$

For the second term, we use

$$\left\| \frac{1}{K} \sum_{k'=1}^K U_{t'}^{(k')} (V_{t'}^{(k')})^\top \right\|_F \leq \frac{1}{K} \sum_{k'=1}^K \|U_{t'}^{(k')} (V_{t'}^{(k')})^\top\|_F \leq \sqrt{r},$$

hence

$$\left\| \sum_{t'=s_t\tau}^{t-1} \frac{1}{K} \sum_{k'=1}^K U_{t'}^{(k')} (V_{t'}^{(k')})^\top \right\|_F \leq \sum_{t'=s_t\tau}^{t-1} \sqrt{r} \leq \tau\sqrt{r}.$$

Substituting these two bounds into (7) yields

$$\frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F \leq \eta \cdot \tau\sqrt{r} + \eta \cdot \tau\sqrt{r} = 2\eta\tau\sqrt{r},$$

which completes the proof. $\square$

LEMMA A.3 (CONSENSUS ERROR OF STORM-REUSE ESTIMATOR). Assume Assumptions 1, 2, and 3. Let $M_t^{(k)}$ be updated by the STORM-reuse rule

$$M_t^{(k)} = \beta M_{t-1}^{(k)} + (1 - \beta) \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) + \beta \left(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}) \right),$$

and define $\bar{M}_t = \frac{1}{K} \sum_{k=1}^K M_t^{(k)}$. Assume that at the beginning of each communication round s (i.e., at $t = s\tau$) the estimators are synchronized: $M_{s\tau}^{(k)} = \bar{M}_{s\tau}$ for all k . Then for any $t \in \{s\tau + 1, \dots, (s+1)\tau\}$,

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\bar{M}_t - M_t^{(k)}\|_F] \leq 8\eta\tau^2 L\sqrt{r} + 4\tau\sigma + 2\tau\delta. \quad (8)$$

PROOF. First, simplify (8) by combining the two stochastic gradient terms at $W_t^{(k)}$:

$$M_t^{(k)} = \beta M_{t-1}^{(k)} + \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \beta \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}). \quad (9)$$

Averaging (9) over k yields

$$\bar{M}_t = \beta \bar{M}_{t-1} + \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(W_t^{(j)}; \xi_t^{(j)}) - \beta \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(W_{t-1}^{(j)}; \xi_t^{(j)}).$$

Subtracting (9) from the above gives, for each k ,

$$\bar{M}_t - M_t^{(k)} = \beta(\bar{M}_{t-1} - M_{t-1}^{(k)}) + \Delta_t^{(k)} - \beta \bar{\Delta}_{t-1}^{(k)}, \quad (10)$$

where

$$\begin{aligned} \Delta_t^{(k)} &:= \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(W_t^{(j)}; \xi_t^{(j)}) - \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}), \\ \bar{\Delta}_{t-1}^{(k)} &:= \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(W_{t-1}^{(j)}; \xi_t^{(j)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}). \end{aligned}$$

Taking Frobenius norms, expectations, and averaging over k ,

$$\begin{aligned} a_t &:= \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\bar{M}_t - M_t^{(k)}\|_F \\ &\leq \beta a_{t-1} + \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\Delta_t^{(k)}\|_F + \beta \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\bar{\Delta}_{t-1}^{(k)}\|_F. \end{aligned} \quad (11)$$

We next bound the two “input” terms. For $\Delta_t^{(k)}$, add and subtract the true gradients:

$$\nabla f^{(j)}(W_t^{(j)}; \xi_t^{(j)}) = \nabla f^{(j)}(W_t^{(j)}) + \zeta_t^{(j)}, \quad \mathbb{E} \|\zeta_t^{(j)}\|_F \leq \sigma,$$

where the variance assumption implies $\mathbb{E} \|\zeta_t^{(j)}\|_F \leq (\mathbb{E} \|\zeta_t^{(j)}\|_F^2)^{1/2} \leq \sigma$. Then by triangle inequality,

$$\begin{aligned} \|\Delta_t^{(k)}\|_F &\leq \left\| \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(W_t^{(j)}) - \nabla f^{(k)}(W_t^{(k)}) \right\|_F + \left\| \frac{1}{K} \sum_{j=1}^K \zeta_t^{(j)} - \zeta_t^{(k)} \right\|_F \\ &\leq \left\| \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(W_t^{(j)}) - \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(\bar{W}_t) \right\|_F + \left\| \frac{1}{K} \sum_{j=1}^K \nabla f^{(j)}(\bar{W}_t) - \nabla f^{(k)}(\bar{W}_t) \right\|_F \end{aligned}$$

$$+ \|\nabla f^{(k)}(\bar{W}_t) - \nabla f^{(k)}(W_t^{(k)})\|_F + \left\| \frac{1}{K} \sum_{j=1}^K \zeta_t^{(j)} - \zeta_t^{(k)} \right\|_F.$$

Using L -smoothness, heterogeneity, and triangle inequality,

$$\|\Delta_t^{(k)}\|_F \leq 2L \cdot \frac{1}{K} \sum_{j=1}^K \|W_t^{(j)} - \bar{W}_t\|_F + \delta + \left\| \frac{1}{K} \sum_{j=1}^K \zeta_t^{(j)} - \zeta_t^{(k)} \right\|_F.$$

Taking expectations and averaging over k ,

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\Delta_t^{(k)}\|_F \leq 2L \cdot \frac{1}{K} \sum_{j=1}^K \mathbb{E} \|W_t^{(j)} - \bar{W}_t\|_F + \delta + \frac{1}{K} \sum_{k=1}^K \mathbb{E} \left\| \frac{1}{K} \sum_{j=1}^K \zeta_t^{(j)} - \zeta_t^{(k)} \right\|_F.$$

Finally, by triangle inequality and $\mathbb{E} \|\zeta_t^{(j)}\|_F \leq \sigma$,

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \left\| \frac{1}{K} \sum_{j=1}^K \zeta_t^{(j)} - \zeta_t^{(k)} \right\|_F \leq \mathbb{E} \left\| \frac{1}{K} \sum_{j=1}^K \zeta_t^{(j)} \right\|_F + \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\zeta_t^{(k)}\|_F \leq \sigma + \sigma = 2\sigma.$$

Hence,

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\Delta_t^{(k)}\|_F \leq 2L \cdot \frac{1}{K} \sum_{j=1}^K \mathbb{E} \|W_t^{(j)} - \bar{W}_t\|_F + 2\sigma + \delta. \quad (12)$$

The same bound holds for $\tilde{\Delta}_{t-1}^{(k)}$ with t replaced by $t-1$.

Substituting (12) (and its $t-1$ analogue) into (11) gives

$$a_t \leq \beta a_{t-1} + (1+\beta) \left(2L \cdot \frac{1}{K} \sum_{j=1}^K \mathbb{E} \|W_t^{(j)} - \bar{W}_t\|_F + 2\sigma + \delta \right).$$

Using the model-consensus bound (Lemma A.2), $\frac{1}{K} \sum_{j=1}^K \mathbb{E} \|W_t^{(j)} - \bar{W}_t\|_F \leq 2\eta\tau\sqrt{r}$, we obtain

$$a_t \leq \beta a_{t-1} + (1+\beta) (4\eta\tau L\sqrt{r} + 2\sigma + \delta). \quad (13)$$

Now fix a communication round s and consider $t \in \{s\tau + 1, \dots, (s+1)\tau\}$. By the synchronization assumption, $a_{s\tau} = 0$. Unrolling (13) yields

$$a_t \leq (1+\beta) (4\eta\tau L\sqrt{r} + 2\sigma + \delta) \sum_{i=0}^{t-(s\tau+1)} \beta^i.$$

Finally, $\sum_{i=0}^m \beta^i \leq \min\{\tau, 1/(1-\beta)\}$ for $m \leq \tau-1$ gives

$$\begin{aligned} \frac{1}{K} \sum_{k=1}^K \mathbb{E} [\|\bar{M}_t - M_t^{(k)}\|_F] &\leq (1+\beta) (4\eta\tau L\sqrt{r} + 2\sigma + \delta) \cdot \min\left\{\tau, \frac{1}{1-\beta}\right\} \\ &\leq 8\eta\tau^2 L\sqrt{r} + 4\tau\sigma + 2\tau\delta. \end{aligned}$$

This completes the proof. $\square$

LEMMA A.4 (STORM-TYPE AVERAGED ESTIMATOR ERROR). Suppose each worker k maintains $M_t^{(k)}$ via $M_t^{(k)} = \beta M_{t-1}^{(k)} + (1-\beta) \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) + \beta(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}))$. Assume assumption 1, 2 and 3. Define $\bar{M}_t := \frac{1}{K} \sum_{k=1}^K M_t^{(k)}$, and $\bar{a}_t := \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \bar{M}_t \right\|_F^2$. Given $\bar{a}_0 \leq \sigma^2/(KB_0)$, for any $T \geq 1$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \bar{M}_t \right\|_F \leq \sqrt{\frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{8L^2\eta^2r}{1-\beta}}.$$

PROOF. From $M_t^{(k)} = \beta M_{t-1}^{(k)} + (1-\beta) \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) + \beta(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}))$, subtract $\nabla f^{(k)}(W_t^{(k)})$ on both sides and average over k . Using $\bar{M}_t = \frac{1}{K} \sum_{k=1}^K M_t^{(k)}$, we obtain

$$\begin{aligned} \bar{M}_t - \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) &= \beta \left(\bar{M}_{t-1} - \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_{t-1}^{(k)}) \right) \\ &\quad + (1-\beta) \cdot \frac{1}{K} \sum_{k=1}^K \left(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_t^{(k)}) \right) \end{aligned}$$

$$+ \beta \cdot \frac{1}{K} \sum_{k=1}^K \left[(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)})) - (\nabla f^{(k)}(W_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)})) \right]. \quad (14)$$

Take squared Frobenius norms and use $\|x + y + z\|_F^2 \leq 3(\|x\|_F^2 + \|y\|_F^2 + \|z\|_F^2)$, then expectation:

$$\begin{aligned} \bar{a}_t &\leq 3\beta^2 \bar{a}_{t-1} + 3(1-\beta)^2 \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K (\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_t^{(k)})) \right\|_F^2 \\ &\quad + 3\beta^2 \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \left[(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)})) - (\nabla f^{(k)}(W_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)})) \right] \right\|_F^2. \end{aligned} \quad (15)$$

For the middle term in (15), by independence across workers,

$$\begin{aligned} &\mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K (\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_t^{(k)})) \right\|_F^2 \\ &= \frac{1}{K^2} \sum_{k=1}^K \mathbb{E} \left\| \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_t^{(k)}) \right\|_F^2 \\ &\leq \frac{\sigma^2}{K}. \end{aligned}$$

For the last term in (15), use Jensen:

$$\mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K Z_k \right\|_F^2 \leq \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|Z_k\|_F^2,$$

where Z_k denotes the bracketed expression for worker k . For each k , apply $\|A - B\|_F^2 \leq 2\|A\|_F^2 + 2\|B\|_F^2$, then we have

$$\begin{aligned} &\mathbb{E} \left\| (\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)})) - (\nabla f^{(k)}(W_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)})) \right\|_F^2 \\ &\leq 2\mathbb{E} \left\| \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}) \right\|_F^2 + 2\mathbb{E} \left\| \nabla f^{(k)}(W_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}) \right\|_F^2 \\ &\leq 2L^2 \|W_t^{(k)} - W_{t-1}^{(k)}\|_F^2 + 2L^2 \|W_t^{(k)} - W_{t-1}^{(k)}\|_F^2 = 4L^2 \|W_t^{(k)} - W_{t-1}^{(k)}\|_F^2. \end{aligned}$$

Using the update rule,

$$\|W_t^{(k)} - W_{t-1}^{(k)}\|_F^2 = \eta^2 \|U_{t-1}^{(k)}(V_{t-1}^{(k)})^\top\|_F^2 \leq \eta^2 r,$$

so the last term in (15) is bounded by $4L^2\eta^2 r$.

Substituting the two bounds into (15) yields

$$\bar{a}_t \leq 3\beta^2 \bar{a}_{t-1} + \frac{3(1-\beta)^2 \sigma^2}{K} + 12\beta^2 L^2 \eta^2 r.$$

Absorbing numerical constants (and using $\beta^2 \leq 1$) gives

$$\bar{a}_t \leq \beta^2 \bar{a}_{t-1} + \frac{2(1-\beta)^2 \sigma^2}{K} + 8L^2 \eta^2 r.$$

Unrolling it and using $\sum_{j=0}^{t-1} \beta^{2j} \leq \frac{1}{1-\beta}$ gives

$$\frac{1}{T} \sum_{t=0}^{T-1} \bar{a}_t \leq \frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{8L^2\eta^2 r}{1-\beta}.$$

Finally, Jensen implies

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \bar{M}_t \right\|_F \leq \sqrt{\frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{8L^2\eta^2 r}{1-\beta}}.$$

□

THEOREM A.5. Suppose Assumption 1,4,2 and 3 hold. If we apply Algorithm 1 with $\eta_t = \eta$, then, the averaged gradient norm satisfies, for any $T \geq 1$

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F &\leq \frac{\mathbb{E} f(\bar{W}_0) - f_*}{\eta T} + 16\tau^2 Lr + 8\sqrt{r}\tau\sigma + 4\sqrt{r}\tau\delta + 4\eta\tau Lr + \frac{\eta r L}{2} \\ &\quad + 2\sqrt{r} \sqrt{\frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{2L^2\eta^2 r}{1-\beta}}. \end{aligned}$$

Let $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1-\beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, if $T > K$, then we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F = O(\frac{1}{K^{1/3}T^{1/3}}).$$

PROOF. Using lemma A.1, we have

$$\begin{aligned} \mathbb{E} f(\bar{W}_{t+1}) &\leq \mathbb{E} f(\bar{W}_t) - \eta \mathbb{E} \|\nabla f(\bar{W}_t)\|_F \\ &\quad + 2\eta\sqrt{r} \frac{L}{K} \sum_{k=1}^K \mathbb{E} \|\bar{W}_t - W_t^{(k)}\|_F \\ &\quad + 2\eta\sqrt{r} \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F \\ &\quad + 2\eta\sqrt{r} \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\bar{M}_t - M_t^{(k)}\|_F + \frac{\eta^2 r L}{2}. \end{aligned} \tag{16}$$

We bound each of the three additional terms.

For the model consensus term, Lemma A.2 implies

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\bar{W}_t - W_t^{(k)}\|_F \leq 2\eta\tau\sqrt{r}.$$

Thus,

$$2\eta\sqrt{r} \frac{L}{K} \sum_{k=1}^K \mathbb{E} \|\bar{W}_t - W_t^{(k)}\|_F \leq 2\eta\sqrt{r} \cdot L \cdot 2\eta\tau\sqrt{r} = 4\eta^2\tau Lr.$$

For the estimator consensus term, Lemma A.3 gives

$$\begin{aligned} 2\eta\sqrt{r} \frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\bar{M}_t - M_t^{(k)}\|_F &\leq 2\eta\sqrt{r} (8\eta\tau^2 L\sqrt{r} + 4\tau\sigma + 2\tau\delta) \\ &= 16\eta^2\tau^2 Lr + 8\eta\sqrt{r}\tau\sigma + 4\eta\sqrt{r}\tau\delta. \end{aligned}$$

Substituting these three bounds into (16), we obtain

$$\begin{aligned} \mathbb{E} f(\bar{W}_{t+1}) &\leq \mathbb{E} f(\bar{W}_t) - \eta \mathbb{E} \|\nabla f(\bar{W}_t)\|_F \\ &\quad + 4\eta^2\tau Lr + 2\eta\sqrt{r} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F \\ &\quad + 16\eta^2\tau^2 Lr + 8\eta\sqrt{r}\tau\sigma + 4\eta\sqrt{r}\tau\delta \\ &\quad + \frac{\eta^2 r L}{2}. \end{aligned}$$

Rearranging terms, we get

$$\begin{aligned} \eta \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F &\leq \mathbb{E} f(\bar{W}_0) - \mathbb{E} f(\bar{W}_T) + 16\eta^2\tau^2 LrT + 8\eta\sqrt{r}\tau\sigma T \\ &\quad + 4\eta\sqrt{r}\tau\delta T + 4\eta^2\tau LrT + \frac{\eta^2 r L}{2} T \\ &\quad + 2\eta\sqrt{r} \sum_{t=0}^{T-1} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F. \end{aligned}$$

Dividing by T and using Jensen's inequality, $f(\bar{W}_T) \geq f_*$, we have that

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F &\leq \frac{\mathbb{E} f(\bar{W}_0) - f_*}{\eta T} + 16\eta\tau^2 Lr + 8\sqrt{r}\tau\sigma \\ &\quad + 4\sqrt{r}\tau\delta + 4\eta\tau Lr + \frac{\eta r L}{2} \\ &\quad + 2\sqrt{r} \sqrt{\frac{\sigma^2}{KB_0(1-\beta)T} + \frac{2(1-\beta)\sigma^2}{K} + \frac{2L^2\eta^2 r}{1-\beta}}. \end{aligned}$$

Let $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1-\beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, if $T > K$, then we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_F = O(\frac{1}{K^{1/3}T^{1/3}}).$$

□

B Proof of theorem 5.3

LEMMA B.1. *Suppose assumption 4,2 and 3 hold. Let $f(W) = \frac{1}{K} \sum_{k=1}^K f^{(k)}(W)$, $\nabla f(W) = \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W)$, then the following descent inequality holds:*

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta(1 - \varepsilon_q) \|\nabla f(\bar{W}_t)\|_* \\ &\quad + 2\eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} + 2\eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* + \frac{\eta^2 L}{2} (1 + \varepsilon_q)^2 \\ &\quad + 2\eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \end{aligned}$$

PROOF. First, Let P_t be the polar factor of M_t , i.e., $P_t = \text{Polar}(M_t)$. Hence, $P_t : \text{Polar}(M_t) = U_t V_t^\top$. By Assumption 1, since $W_t^{(k)} = W_{t-1}^{(k)} - \eta O_t^{(k)}$, we have $\bar{W}_t = \bar{W}_{t-1} - \eta \frac{1}{K} \sum_{k=1}^K O_t^{(k)}$

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) + \langle \nabla f(\bar{W}_t), \bar{W}_{t+1} - \bar{W}_t \rangle + \frac{L}{2} \|\bar{W}_{t+1} - \bar{W}_t\|_{\text{op}}^2 \\ &\leq f(\bar{W}_t) - \eta \langle \nabla f(\bar{W}_t), \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \rangle + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \\ &= f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \langle \nabla f(\bar{W}_t) - \bar{M}_t, O_t^{(k)} \rangle - \eta \frac{1}{K} \sum_{k=1}^K \langle \bar{M}_t - M_t^{(k)}, O_t^{(k)} \rangle \\ &\quad - \eta \frac{1}{K} \sum_{k=1}^K \langle M_t^{(k)}, O_t^{(k)} \rangle + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \\ &\leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \langle M_t^{(k)}, U_t^{(k)} V_t^{(k)\top} \rangle - \eta \frac{1}{K} \sum_{k=1}^K \langle M_t^{(k)}, U_t^{(k)} V_t^{(k)\top} - O_t^{(k)} \rangle \\ &\quad + \eta \frac{1}{K} \sum_{k=1}^K \|\nabla f(\bar{W}_t) - \bar{M}_t\|_* \|O_t^{(k)}\|_{\text{op}} \\ &\quad + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \|O_t^{(k)}\|_{\text{op}} + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \\ &\leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* + \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* \|U_t^{(k)} V_t^{(k)\top} - O_t^{(k)}\|_{\text{op}} \\ &\quad + \eta \frac{1}{K} \sum_{k=1}^K \|\nabla f(\bar{W}_t) - \bar{M}_t\|_* \|O_t^{(k)}\|_{\text{op}} \\ &\quad + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \|O_t^{(k)}\|_{\text{op}} + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \end{aligned}$$

$$\begin{aligned}
& \leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* + \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* \|U_t^{(k)} V_t^{(k)\top} - O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \|\nabla f(\bar{W}_t) - \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)})\|_* \|O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \|O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \|O_t^{(k)}\|_{\text{op}} + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \\
& \leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* + \eta \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)}\|_* \|U_t^{(k)} V_t^{(k)\top} - O_t^{(k)}\|_{\text{op}} \\
& + \eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \|O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \|O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \|O_t^{(k)}\|_{\text{op}} + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \tag{17}
\end{aligned}$$

Now, we define the polar approximation error, which is the error between the exact polar factor $P_t^{(k)} = \text{Polar}(M_t^{(k)}) = U_t^{(k)} V_t^{(k)\top}$ and the actual step $O_t^{(k)}$ generated by q -step Newton-Schulz steps, i.e., $O_t^{(k)} = \text{Newton-Schulz}(M_{t,q}^{(k)})$, measured in the operator norm.

$$\varepsilon_{t,q}^{(k)} := \|O_t^{(k)} - U_t^{(k)} V_t^{(k)\top}\|_{\text{op}} \quad \text{and} \quad \varepsilon_q := \sup_t \varepsilon_{t,q}^{(k)}$$

Since $\|U_t^{(k)} V_t^{(k)\top}\|_{\text{op}} \leq 1$, $\|O_t^{(k)}\|_{\text{op}} \leq 1 + \varepsilon_{t,q}^{(k)} \leq 1 + \varepsilon_q$, we have

$$\begin{aligned}
f(\bar{W}_{t+1}) & \leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \left(1 - \|U_t^{(k)} V_t^{(k)\top} - O_t^{(k)}\|_{\text{op}} \right) \|M_t^{(k)}\|_* \\
& + \eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \|O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \|O_t^{(k)}\|_{\text{op}} \\
& + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \|O_t^{(k)}\|_{\text{op}} + \frac{\eta^2 L}{2} \left\| \frac{1}{K} \sum_{k=1}^K O_t^{(k)} \right\|_{\text{op}}^2 \\
& \leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \left(1 - \varepsilon_{t,q}^{(k)} \right) \|M_t^{(k)}\|_* \\
& + \eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \left(1 + \varepsilon_{t,q}^{(k)} \right) \\
& + \eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \left(1 + \varepsilon_{t,q}^{(k)} \right) \\
& + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \left(1 + \varepsilon_{t,q}^{(k)} \right) + \frac{\eta^2 L}{2} \frac{1}{K} \sum_{k=1}^K \left(1 + \varepsilon_{t,q}^{(k)} \right)^2 \\
& \leq f(\bar{W}_t) - \eta \frac{1}{K} \sum_{k=1}^K \left(1 - \varepsilon_{t,q}^{(k)} \right) \|\nabla f(\bar{W}_t)\|_*
\end{aligned}$$

$$\begin{aligned}
& + 2\eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \\
& + 2\eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \\
& + \eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* \left(1 + \varepsilon_{t,q}^{(k)}\right) + \frac{\eta^2 L}{2} \frac{1}{K} \sum_{k=1}^K \left(1 + \varepsilon_{t,q}^{(k)}\right)^2
\end{aligned} \tag{18}$$

Here $(1 - \varepsilon_{t,q}^{(k)}) + (1 + \varepsilon_{t,q}^{(k)}) = 2$. Since $\varepsilon_{t,q}^{(k)} \leq \varepsilon_q$, we have

$$\begin{aligned}
f(\bar{W}_{t+1}) & \leq f(\bar{W}_t) - \eta (1 - \varepsilon_q) \|\nabla f(\bar{W}_t)\|_* \\
& + 2\eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} + 2\eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* + \frac{\eta^2 L}{2} (1 + \varepsilon_q)^2 \\
& + 2\eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_*
\end{aligned}$$

□

LEMMA B.2. Suppose Assumption 4.2 and 3 hold. At the beginning of each communication period of length τ , the models satisfy the synchronization condition $W_{s_t\tau}^{(k)} = \bar{W}_{s_t\tau}, \forall k, \forall s \in \mathbb{N}$. Then for any $t \geq 0$, we have

$$\frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \leq 2\eta\tau(1 + \varepsilon_q).$$

PROOF. Let $s_t = \lfloor t/\tau \rfloor$. Unrolling the updates from the beginning of the current communication round,

$$W_t^{(k)} = W_{s_t\tau}^{(k)} - \eta \sum_{t'=s_t\tau}^{t-1} O_{t'}^{(k')}, \quad \bar{W}_t = \bar{W}_{s_t\tau} - \eta \sum_{t'=s_t\tau}^{t-1} \frac{1}{K} \sum_{k'=1}^K O_{t'}^{(k')}.$$

By the synchronization condition $W_{s_t\tau}^{(k)} = \bar{W}_{s_t\tau}$, we obtain

$$\bar{W}_t - W_t^{(k)} = -\eta \sum_{t'=s_t\tau}^{t-1} \left(\frac{1}{K} \sum_{k'=1}^K O_{t'}^{(k')} - O_{t'}^{(k)} \right).$$

Thus,

$$\begin{aligned}
\frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} & = \eta \frac{1}{K} \sum_{k=1}^K \left\| \sum_{t'=s_t\tau}^{t-1} \left(O_{t'}^{(k)} - \frac{1}{K} \sum_{k'=1}^K O_{t'}^{(k')} \right) \right\|_{\text{op}} \\
& \leq \eta \frac{1}{K} \sum_{k=1}^K \left\| \sum_{t'=s_t\tau}^{t-1} O_{t'}^{(k)} \right\|_{\text{op}} \\
& \quad + \eta \frac{1}{K} \sum_{k=1}^K \left\| \sum_{t'=s_t\tau}^{t-1} \frac{1}{K} \sum_{k'=1}^K O_{t'}^{(k')} \right\|_{\text{op}}.
\end{aligned} \tag{19}$$

For the first term, using the triangle inequality and $\|O_t^{(k)}\|_{\text{op}} \leq (1 + \varepsilon_{t,q}^{(k)}) \leq (1 + \varepsilon_q)$,

$$\left\| \sum_{t'=s_t\tau}^{t-1} O_{t'}^{(k)} \right\|_{\text{op}} \leq \sum_{t'=s_t\tau}^{t-1} \|O_{t'}^{(k)}\|_{\text{op}} \leq (t - s_t\tau)(1 + \varepsilon_q) \leq \tau(1 + \varepsilon_q).$$

For the second term, we use

$$\left\| \frac{1}{K} \sum_{k'=1}^K O_{t'}^{(k')} \right\|_{\text{op}} \leq \frac{1}{K} \sum_{k'=1}^K \|O_{t'}^{(k')}\|_{\text{op}} \leq (1 + \varepsilon_q),$$

hence

$$\left\| \sum_{t'=s_t\tau}^{t-1} \frac{1}{K} \sum_{k'=1}^K O_{t'}^{(k')} \right\|_{\text{op}} \leq \sum_{t'=s_t\tau}^{t-1} (1 + \varepsilon_q) \leq \tau(1 + \varepsilon_q).$$

Substituting these two bounds into (19) yields

$$\frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \leq \eta \cdot \tau(1 + \varepsilon_q) + \eta \cdot \tau(1 + \varepsilon_q) = 2\eta\tau(1 + \varepsilon_q),$$

which completes the proof. $\square$

LEMMA B.3. Assume Assumptions 4, 2, and 3. Let $M_t^{(k)}$ be updated by the STORM-reuse rule

$$M_t^{(k)} = \beta M_{t-1}^{(k)} + (1 - \beta) \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) + \beta \left(\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}) \right),$$

and define $\bar{M}_t = \frac{1}{K} \sum_{k=1}^K M_t^{(k)}$. Assume that at the beginning of each communication round s (i.e., at $t = s\tau$) the estimators are synchronized: $M_{s\tau}^{(k)} = \bar{M}_{s\tau}$ for all k . Then for any $t \in \{s\tau + 1, \dots, (s+1)\tau\}$,

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\bar{M}_t - M_t^{(k)}\|_*] \leq 8\eta\tau^2 L(1 + \varepsilon_q) + 4\tau\sqrt{r}\sigma + 2\tau\sqrt{r}\delta.$$

PROOF. The proof follows the same idea as lemma A.3. By using the new update rules $W_t^{(k)} = W_{t-1}^{(k)} - \eta O_t^{(k)}$, $\|O_t^{(k)}\|_{\text{op}} \leq (1 + \varepsilon_{t,q}^{(k)}) \leq (1 + \varepsilon_q)$ and $\|W\|_* \leq \sqrt{r}\|W\|_F$, we have that

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E}[\|\bar{M}_t - M_t^{(k)}\|_*] \leq 8\eta\tau^2 L(1 + \varepsilon_q) + 4\tau\sqrt{r}\sigma + 2\tau\sqrt{r}\delta.$$

$\square$

LEMMA B.4. Suppose each worker k maintains $M_t^{(k)}$ via $M_t^{(k)} = \beta M_{t-1}^{(k)} + (1 - \beta) \nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) + \beta (\nabla f^{(k)}(W_t^{(k)}; \xi_t^{(k)}) - \nabla f^{(k)}(W_{t-1}^{(k)}; \xi_t^{(k)}))$. Assume assumption 4, 2 and 3. Define $\bar{M}_t := \frac{1}{K} \sum_{k=1}^K M_t^{(k)}$, and $\bar{a}_t := \mathbb{E}[\frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \bar{M}_t]_F^2$. Given $\bar{a}_0 \leq \sigma^2/(KB_0)$, for any $T \geq 1$,

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \bar{M}_t \right\|_* \leq \sqrt{\frac{r\sigma^2}{KB_0(1 - \beta)T} + \frac{2(1 - \beta)r\sigma^2}{K} + \frac{8L^2\eta^2(1 + \varepsilon_q)^2}{1 - \beta}}.$$

PROOF. The proof follows the same idea as lemma A.4. By using the new update rules $W_t^{(k)} = W_{t-1}^{(k)} - \eta O_t^{(k)}$, $\|O_t^{(k)}\|_{\text{op}} \leq (1 + \varepsilon_{t,q}^{(k)}) \leq (1 + \varepsilon_q)$ and $\|W\|_* \leq \sqrt{r}\|W\|_F$, we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \bar{M}_t \right\|_* \leq \sqrt{\frac{r\sigma^2}{KB_0(1 - \beta)T} + \frac{2(1 - \beta)r\sigma^2}{K} + \frac{8L^2\eta^2(1 + \varepsilon_q)^2}{1 - \beta}}.$$

$\square$

THEOREM B.5. Suppose Assumption 4,2 and 3 hold. Consider applying Algorithm 3 with a constant stepsize $\eta_t = \eta$. Assume that there exists a constant $\chi_q = \frac{1}{1 - \varepsilon_q} > 0$, depending only on the number q of Newton-Schulz steps, such that for any $T \geq 1$,

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_* &\leq \chi_q \left(\frac{\mathbb{E} f(\bar{W}_0) - f_*}{\eta T} + 16\eta\tau^2 L(1 + \varepsilon_q) + 8\sqrt{r}\tau\sigma \right. \\ &\quad \left. + 4\sqrt{r}\tau\delta + 4\eta\tau L(1 + \varepsilon_q) + \frac{\eta(1 + \varepsilon_q)^2 L}{2} \right. \\ &\quad \left. + 2\sqrt{\frac{r\sigma^2}{KB_0(1 - \beta)T} + \frac{2r(1 - \beta)\sigma^2}{K} + \frac{2L^2\eta^2(1 + \varepsilon_q)^2}{1 - \beta}} \right). \end{aligned}$$

Let $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1 - \beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, if $T > K$, then we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(\bar{W}_t)\|_* = \chi_q O\left(\frac{1}{K^{1/3}T^{1/3}}\right).$$

PROOF. Using lemma B.1, we have

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta(1 - \varepsilon_q) \|\nabla f(\bar{W}_t)\|_* \\ &\quad + 2\eta L \frac{1}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} + 2\eta \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_* + \frac{\eta^2 L}{2} (1 + \varepsilon_q)^2 \end{aligned}$$

$$+ 2\eta \frac{1}{K} \sum_{k=1}^K \left\| \frac{1}{K} \sum_{k=1}^K f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \quad (20)$$

We bound each of the three additional terms.

For the model consensus term, Lemma B.2 implies

$$\frac{1}{K} \sum_{k=1}^K \mathbb{E} \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \leq 2\eta\tau(1 + \varepsilon_q).$$

Thus,

$$2\eta \frac{L}{K} \sum_{k=1}^K \mathbb{E} \|\bar{W}_t - W_t^{(k)}\|_{\text{op}} \leq 2\eta \cdot L \cdot 2\eta\tau(1 + \varepsilon_q) = 4\eta^2\tau L(1 + \varepsilon_q).$$

For the estimator consensus term, Lemma B.3 gives

$$2\eta \frac{1}{K} \sum_{k=1}^K \mathbb{E} [\|\bar{M}_t - M_t^{(k)}\|_*] \leq 16\eta^2\tau^2 L(1 + \varepsilon_q) + 8\eta\tau\sqrt{r}\sigma + 4\eta\tau\sqrt{r}\delta.$$

Substituting these three bounds into (20), we obtain

$$\begin{aligned} \mathbb{E}f(\bar{W}_{t+1}) &\leq \mathbb{E}f(\bar{W}_t) - \eta(1 - \varepsilon_q) \mathbb{E}\|\nabla f(\bar{W}_t)\|_* \\ &\quad + 4\eta^2\tau L(1 + \varepsilon_q) + 2\eta \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_* \\ &\quad + 16\eta^2\tau^2 L(1 + \varepsilon_q) + 8\eta\tau\sqrt{r}\sigma + 4\eta\tau\sqrt{r}\delta \\ &\quad + \frac{\eta^2 L}{2} (1 + \varepsilon_q)^2. \end{aligned}$$

Rearranging terms, we get

$$\begin{aligned} \eta(1 - \varepsilon_q) \sum_{t=0}^{T-1} \mathbb{E}\|\nabla f(\bar{W}_t)\|_F &\leq \mathbb{E}f(\bar{W}_0) - \mathbb{E}f(\bar{W}_T) + 16\eta^2\tau^2 L(1 + \varepsilon_q)T + 8\eta\tau\sqrt{r}\sigma T \\ &\quad + 4\eta\tau\sqrt{r}\delta T + 4\eta^2\tau L(1 + \varepsilon_q)T + \frac{\eta^2(1 + \varepsilon_q)^2 L}{2} T \\ &\quad + 2\eta \sum_{t=0}^{T-1} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F. \end{aligned}$$

Dividing by T and using Jensen's inequality, $f(\bar{W}_T) \geq f_*$, we have that

$$\begin{aligned} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla f(\bar{W}_t)\|_* &\leq \frac{1}{(1 - \varepsilon_q)} \left(\frac{\mathbb{E}f(\bar{W}_0) - f_*}{\eta T} + 16\eta\tau^2 L(1 + \varepsilon_q) + 8\sqrt{r}\tau\sigma \right. \\ &\quad \left. + 4\sqrt{r}\tau\delta + 4\eta\tau L(1 + \varepsilon_q) + \frac{\eta(1 + \varepsilon_q)^2 L}{2} \right. \\ &\quad \left. + 2\sqrt{\frac{r\sigma^2}{KB_0(1 - \beta)T} + \frac{2r(1 - \beta)\sigma^2}{K} + \frac{2L^2\eta^2(1 + \varepsilon_q)^2}{1 - \beta}} \right). \end{aligned}$$

Let $\eta = O(\frac{K^{1/3}}{T^{2/3}})$, $1 - \beta = O(\frac{K^{1/3}}{T^{2/3}})$, $\tau = O(\frac{1}{K^{1/3}T^{1/3}})$, $B_0 = O(T^{1/3})$, if $T > K$, then we have that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}\|\nabla f(\bar{W}_t)\|_* = \frac{1}{(1 - \varepsilon_q)} O\left(\frac{1}{K^{1/3}T^{1/3}}\right).$$

By defining $\chi_q = \frac{1}{(1 - \varepsilon_q)}$, we complete the proof. $\square$

C Proof of Theorem 6.1

Proof sketch for the truncated case (relative to the full- M proof). The proof follows exactly the same steps as the full- M case. The only difference is that the algorithm uses rank- r truncated matrices, denoted by

$$M_{t,r}^{(k)} := [M_t^{(k)}]_r, \quad \bar{M}_{t,r} := \frac{1}{K} \sum_{k=1}^K M_{t,r}^{(k)}.$$

Accordingly, whenever the full- M proof contains terms involving $\bar{M}_t$ and $M_t^{(k)}$, we insert and subtract their truncated counterparts to keep the argument consistent with the rank- r implementation.

(i) Heterogeneity term. In the full- M proof, the heterogeneity term appears as $\|\bar{M}_t - M_t^{(k)}\|_F$. In the truncated case, we decompose it via

$$\bar{M}_t - M_t^{(k)} = (\bar{M}_{t,r} - M_{t,r}^{(k)}) + (\bar{M}_t - \bar{M}_{t,r}) - (M_t^{(k)} - M_{t,r}^{(k)}),$$

and apply the triangle inequality to obtain

$$\|\bar{M}_t - M_t^{(k)}\|_F \leq \|\bar{M}_{t,r} - M_{t,r}^{(k)}\|_F + \|\bar{M}_t - \bar{M}_{t,r}\|_F + \|M_t^{(k)} - M_{t,r}^{(k)}\|_F. \quad (21)$$

Compared with the full- M case, this introduces two additional truncation-error terms: (i) the average truncation bias $\|\bar{M}_t - \bar{M}_{t,r}\|_F$, and (ii) the local truncation residual $\|M_t^{(k)} - M_{t,r}^{(k)}\|_F$.

(ii) Gradient-model mismatch term. In the full- M proof, we bound the mismatch between local true gradients and the surrogate matrices by

$$\left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F.$$

In the truncated case, we add and subtract $\frac{1}{K} \sum_k M_{t,r}^{(k)}$:

$$\frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} = \left(\frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_{t,r}^{(k)} \right) + \left(\frac{1}{K} \sum_{k=1}^K (M_{t,r}^{(k)} - M_t^{(k)}) \right).$$

Applying the triangle inequality and Jensen's inequality yields

$$\left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F \leq \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_{t,r}^{(k)} \right\|_F + \frac{1}{K} \sum_{k=1}^K \|M_{t,r}^{(k)} - M_t^{(k)}\|_F. \quad (22)$$

Thus, relative to the full- M proof, the truncated proof only adds the averaged truncation residual $\frac{1}{K} \sum_k \|M_{t,r}^{(k)} - M_t^{(k)}\|_F$.

(iii) Exact characterization of the truncated descent term. Let

$$A := [U_t^{(k)}]_r [V_t^{(k)}]_r^\top, \quad B := U_t^{(k)} (V_t^{(k)})^\top.$$

Then the following identity holds:

$$\langle M_t^{(k)}, A \rangle = \langle M_t^{(k)}, B \rangle + \langle M_t^{(k)}, A - B \rangle.$$

If $U_t^{(k)}, V_t^{(k)}$ are the left and right singular vectors from the full SVD of $M_t^{(k)}$, i.e., $M_t^{(k)} = U_t^{(k)} \Sigma_t^{(k)} (V_t^{(k)})^\top$, then

$$\langle M_t^{(k)}, B \rangle = \|M_t^{(k)}\|_*.$$

Therefore,

$$\langle M_t^{(k)}, A \rangle = \|M_t^{(k)}\|_* + \langle M_t^{(k)}, A - B \rangle.$$

$$\langle M_t^{(k)}, A - B \rangle = - \sum_{i=r+1}^n \sigma_i,$$

Combining the above bounds yields

$$\langle M_t^{(k)}, A \rangle = \|M_t^{(k)}\|_* - \sum_{i=r+1}^n \sigma_i.$$

Explicit values of the truncation terms. Fix a worker k and write the (full) SVD

$$M_t^{(k)} = U_t^{(k)} \Sigma_t^{(k)} (V_t^{(k)})^\top, \quad \Sigma_t^{(k)} = \text{diag}(\sigma_1^{(k)}, \dots, \sigma_d^{(k)}),$$

with $\sigma_1^{(k)} \geq \dots \geq \sigma_d^{(k)} \geq 0$ and $d = \min\{m, n\}$. Let the rank- r truncated matrix be

$$M_{t,r}^{(k)} := [M_t^{(k)}]_r := U_{t,r}^{(k)} \Sigma_{t,r}^{(k)} (V_{t,r}^{(k)})^\top.$$

Then the truncation residual has the exact Frobenius norm

$$\|M_t^{(k)} - M_{t,r}^{(k)}\|_F = \left(\sum_{i=r+1}^d (\sigma_i^{(k)})^2 \right)^{1/2}.$$

This is the Eckart–Young–Mirsky result specialized to the Frobenius norm.

For the *average truncation bias*, with

$$\bar{M}_t := \frac{1}{K} \sum_{k=1}^K M_t^{(k)}, \quad \bar{M}_{t,r} := \frac{1}{K} \sum_{k=1}^K M_{t,r}^{(k)},$$

we have the exact identity

$$\|\bar{M}_t - \bar{M}_{t,r}\|_F = \left\| \frac{1}{K} \sum_{k=1}^K (M_t^{(k)} - M_{t,r}^{(k)}) \right\|_F.$$

In general, this does *not* reduce to a closed form involving only the singular values $\{\sigma_i^{(k)}\}$, because it depends on how the tail components from different workers align. A standard bound (often sufficient in proofs) is

$$\|\bar{M}_t - \bar{M}_{t,r}\|_F \leq \frac{1}{K} \sum_{k=1}^K \|M_t^{(k)} - M_{t,r}^{(k)}\|_F = \frac{1}{K} \sum_{k=1}^K \left(\sum_{i=r+1}^d (\sigma_i^{(k)})^2 \right)^{1/2}.$$

$$\rho_t := \max_{k \in [K]} \|M_t^{(k)} - M_{t,r}^{(k)}\|_F = \max_{k \in [K]} \left(\sum_{i=r+1}^d (\sigma_i^{(k)})^2 \right)^{1/2}, \quad d = \min\{m, n\}.$$

Define the tail nuclear mass and tail Frobenius residual by

$$\tau_t := \max_{k \in [K]} \sum_{i=r+1}^d \sigma_i(M_t^{(k)}), \quad \rho_t := \max_{k \in [K]} \left(\sum_{i=r+1}^d (\sigma_i^{(k)})^2 \right)^{1/2} = \max_{k \in [K]} \|M_t^{(k)} - M_{t,r}^{(k)}\|_F.$$

By Cauchy–Schwarz,

$$\sum_{i=r+1}^d \sigma_i(M_t^{(k)}) \leq \sqrt{d-r} \left(\sum_{i=r+1}^d (\sigma_i^{(k)})^2 \right)^{1/2},$$

hence $\tau_t \leq \sqrt{d-r} \rho_t$.

$$\begin{aligned} f(\bar{W}_{t+1}) &\leq f(\bar{W}_t) - \eta \|\nabla f(\bar{W}_t)\|_F + 2\eta\sqrt{r} \frac{1}{K} \sum_{k=1}^K \|\bar{M}_t - M_t^{(k)}\|_F + 2\eta\sqrt{r} \frac{L}{K} \sum_{k=1}^K \|\bar{W}_t - W_t^{(k)}\|_F + 2\eta\sqrt{r} \\ &\quad + 2\eta\sqrt{r} \left\| \frac{1}{K} \sum_{k=1}^K \nabla f^{(k)}(W_t^{(k)}) - \frac{1}{K} \sum_{k=1}^K M_t^{(k)} \right\|_F + \eta(6\sqrt{r} + \sqrt{d-r})\rho_t \end{aligned}$$